PERSONALIZED CAR BUYING RECOMMENDATION SYSTEM FOR INDIAN CONSUMERS: LEVERAGING MACHINE LEARNING TO TAILOR PURCHASE DECISIONS

A PROJECT REPORT

Submitted by

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KATTANKULATHUR – 603 203

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ABSTRACT

In India, the demand for personalized auto buying recommendation systems is increasing due to evolving consumer preferences and an expanding array of car models. This research presents an AI-driven recommendation system that integrates comprehensive car data, user preferences, social media sentiment analysis, and financial eligibility to provide tailored suggestions for prospective car buyers. The system employs machine learning techniques to analyze key automotive features like as seating capacity, fuel type, and gearbox, while utilizing user-generated data from social media and reviews to assess public sentiment with NLP methods like DistilBERT. The technology additionally integrates buyer-specific factors such as preferred brands and previous vehicle ownership. Financial analytics evaluate the purchaser's income, existing debt, and qualification for financing alternatives. By integrating these elements, we formulate thorough automobile recommendations that align with client desires while guaranteeing financial viability. This technique streamlines the automobile acquisition process, improves decision-making, and elevates user happiness for both Indian consumers and automotive retailers. Our methodology generated precise suggestions of the top five vehicles tailored to each user's tastes through the application of cosine similarity and weighted scoring. Our objective is to present consumers with the top five vehicles aligned with their preferences, so facilitating informed decision-making.

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LIST OF SYMBOLS AND ABBREVIATIONS

SIAM	Society of Indian Automobile Manufacturers
NLP	Natural Language Processing
EV	Electric Vehicle
SUV	Sports Utility Vehicle
BERT	Bidirectional Encoder Representations from Transformers
GNN	Graph Neural Network
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
EMI	Equated Monthly Installments
CF	Collaborative Filtering
KG	Knowledge Graph
OEM	Original Equipment Manufacturer
OCR	Optical Character Recognition
CNG	Compressed Natural Gas
AMT	Auto Manual Transmission
USS	User Satisfaction Score
PDS	Price Deviation Score
PMR	Preferred Match Rate
FFS	Financial Feasibility Score
WSC	Weighted Score Consistency

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CHAPTER 1

INTRODUCTION

1.1 General

Twenty years ago, buying a car in India was simple. You had a handful of basic models to choose from—mostly small, fuel-efficient, and affordable. Fast forward to today, and the scene is completely different. India is now the world's third-largest car market, thanks to rising incomes, crowded cities, and people wanting more from their vehicles. Walk into a showroom or browse online, and you'll find everything—from tiny hatchbacks to luxury SUVs, petrol and diesel engines to electric cars, and features that feel like they're from a sci-fi movie.

Buyers today aren't just walking into dealerships blindly. They're doing their homework—scouring YouTube for reviews, checking comparison websites, reading real owner experiences, and crunching numbers with loan calculators. But here's the funny thing: even with all this information, picking the right car feels harder than ever. Why? Because now, you're not just choosing a car—you're weighing fuel efficiency against resale value, brand trust against after-sales service, and monthly EMIs against how many dealerships are nearby in case something goes wrong.

And just when you think you've figured it out, the rules change. The government tightens emission norms (hello, BS6), electric cars start making headlines, and safety ratings suddenly matter more. Then there's the pandemic effect—buyers now expect seamless online bookings, contactless test drives, and deals tailored just for them.

The old way of buying a car—flipping through brochures or trusting a salesperson's pitch—doesn't work anymore. Those methods miss what really matters to you: your budget, your driving habits, even that little voice in your head that says, "But do I look good in this car?" And with every car company flooding the market with endless options, it's easy to feel overwhelmed.

What buyers need is a smarter way to cut through the noise. Imagine a tool that doesn't just list cars but actually understands you—your daily commute, your financial limits, even your secret love for sunroofs. Instead of generic rankings, it looks at real owner complaints, social media buzz, and whether you can actually afford the EMI without eating instant noodles for a year.

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But India isn't one market—it's a mix of a hundred different ones. In Mumbai's traffic, a compact hatchback makes sense. In Himachal's hills? You'd want something tougher. And let's not forget, most Indians don't pay upfront—they take loans. So, a great car recommendation isn't just about horsepower or mileage; it's about whether the bank will approve your loan and if the EMI fits your salary.

Bottom line? India's car market is booming, but buyers are stuck in analysis paralysis. Between flashy ads, confusing specs, and FOMO (what if a better model launches next month?), a helping hand that actually gets Indian buyers could be a game-changer. That's exactly what this project is about—building a recommendation system that feels less like a robot and more like a trusted friend who knows cars inside out..

1.2 Purpose

Picking a car in India has become ridiculously hard. With dozens of new models, fuel types, and price ranges, most buyers feel completely lost. You want the perfect balance of features, budget, and practicality – but how? That's where we come in. This project builds a recommendation tool that actually understands Indian buyers. No robotic comparisons or sales pitches – just smart, personalized advice that considers your needs, driving habits, and wallet. Because choosing your next car should be exciting, not exhausting.

Traditionally, Indian car buyers have relied on dealers, friends, or family for guidance in selecting a vehicle. However, these recommendations are often based on limited experience and may not align with a buyer's specific needs. With the rise of digital platforms and online reviews, consumers now have access to more information than ever before. Yet, paradoxically, the sheer volume of data makes it more difficult to filter what's relevant. There's no structured system that consolidates technical specifications, sentiment data, and financial insights into a single, user-friendly recommendation.

This project proposes a machine learning-based car recommendation engine that uses a hybrid approach—blending user preferences, online sentiment analysis, and financial viability—to offer the most relevant car suggestions to individual buyers. The aim is to remove the guesswork and manual effort from the car-buying process by developing a system that understands the user like a personal consultant.

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The project serves several important purposes:

1. **To Provide Customized Recommendations**: Every car buyer is different. A first-time buyer in Bengaluru looking for a budget-friendly, fuel-efficient hatchback has very different needs from a Delhi-based entrepreneur seeking a premium electric SUV. This system will capture those individual preferences and recommend vehicles accordingly.
2. **To Incorporate Financial Eligibility**: Car buying in India is often influenced by EMI structures, down payments, and loan approvals. Finding your dream car means nothing if you can't afford it. Our system checks your income, expenses, and loan eligibility first - only showing vehicles that fit your budget. No more falling in love with cars that break the bank.
3. **To Leverage Social Sentiment**: These days, car buyers heavily rely on reviews and word-of-mouth before making decisions. But who has time to read through pages of customer feedback? Our system scans thousands of real owner experiences using smart analysis tools, then boils down what actual drivers love (or hate) about each model providing user the crowd's honest opinions without the endless scrolling.
4. **To Reduce Cognitive Load and Improve User Experience**: Too many options can leave you stuck in car-buying limbo. That's why we narrow it down to just 5 perfect matches - enough choice to feel good about your decision, but not so many that you'll drown in comparisons.
5. **To Bridge the Gap Between Online Research and Purchase Decision**: Most car buyers today do their homework online, but still feel the need to visit multiple showrooms just to be sure. Our system cuts through this uncertainty by giving you reliable, data-backed recommendations - so you can walk into a dealership already confident about your choice.
6. **To Support Dealerships and Auto Platforms**: While we built this system to help car buyers, it's also a powerful tool for dealerships and online platforms. By offering truly personalized recommendations, businesses can provide better customer experiences while naturally increasing sales - without the pushy sales tactics..

In essence, this project is about changing how Indian buys cars. We're creating a recommendation system that understands what really matters - not just engine specs and features, but how the car fits your life, budget, and even your emotions. It's about taking the stress out of car shopping and helping people make choices they'll feel good about long after driving off the lot.

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1.3 Scope

This project focuses on creating a smarter way for Indians to buy cars. We're building a recommendation tool that truly understands what matters to you - your favorite features, budget limits, and even what other buyers like you are saying. It's designed to take the headache out of car shopping by giving personalized advice that actually makes sense for your life.

India's car market is incredibly diverse, making it one of the most challenging to navigate. On any given street, you might see a compact hatchback squeezing through traffic alongside a luxury SUV - with price tags ranging from ₹4 lakh to ₹10 crore. With more than 40 brands and hundreds of models to choose from, buyers often feel overwhelmed trying to find the right match for their budget and lifestyle.

What makes India unique is how local factors influence car choices. A vehicle perfect for Mumbai's crowded streets might struggle in rural areas with rough roads. Climate affects whether you prioritize strong AC or cold-start performance. Even fuel availability varies across regions. That's why generic recommendations don't work here - our system understands these nuances to give truly personalized advice.

Our system helps everyday Indian car buyers by focusing on vehicles priced below ₹50 lakh. Through smart algorithms, it analyzes your preferences, budget and local conditions to recommend the best matches - taking the guesswork out of car buying. Below are the key dimensions of this scope:

1.3.1 Functional Scope

The functional capabilities of the system include:

- **User Input Collection**: Through a user-friendly interface or widget, the system captures essential details such as budget, seating preference, preferred fuel type, past ownership history, and brand inclination.
- **Technical Filtering**: Based on inputs, the system filters out irrelevant models using structured car specifications like fuel type, transmission mode, mileage, seating capacity, budget and brand
- **Sentiment Analysis**: The system utilizes deep learning models (like DistilBERT) to analyze user reviews and social media feedback to understand public opinion on various car models.

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- **Financial Assessment**: Using user-provided data such as income, existing financial commitments, and desired down payment, the system performs loan eligibility checks and let user aware of their affordability.
- **Recommendation Engine**: Combines all above inputs into a scoring model to generate the top 5 most suitable car recommendations for each user.
- **Feedback Integration**: Future iterations of the system can accept user ratings and satisfaction scores post-recommendation to retrain the model and improve its accuracy over time.

1.3.2 Dataset Scope

The system draws data from multiple sources:

- A curated Kaggle dataset of Indian car specifications (1994 rows × 147 features).
- Web-scraped reviews and sentiment scores from social media platforms
- Financial data gathered from Indian banking norms (e.g., SBI, HDFC car loan eligibility).
- Manually researched service costs, seating capacity, and 2021–2024 monthly car sales trends.

All data is processed and normalized to maintain consistency, accuracy, and completeness.

1.3.3 Technological Scope

This project incorporates a blend of AI, ML, and NLP technologies:

- **Content-based filtering** for matching user preferences with car attributes.
- **Sentiment classification** using deep learning models like DistilBERT.
- **Rule-based financial evaluation** for EMI and loan eligibility analysis.
- **Weighted scoring algorithm** using real survey data to determine importance of features like fuel efficiency, brand preference, and service cost.

The system is built in Python with Jupyter Notebook as the development environment and can be deployed as a web-based or standalone application.

1.3.4 Application Scope

The primary users of this system are individual car buyers in India, especially those seeking:

- A simplified decision-making process.
- Clarity amidst market overload.
- Budget-aware recommendations.

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- Trustworthy guidance based on real consumer sentiment.

However, the system also has broader applications. Automobile dealerships can integrate such a recommendation engine to personalize user experiences and drive conversions. The model also has potential use in academic research, consumer analytics, and financial advising platforms.

1.3.5 Limitations of Scope

While the system is robust, it does have defined boundaries:

- It currently supports only new car purchases, not used or certified pre-owned vehicles.
- The financial eligibility is estimated using simplified rules and may not account for dynamic banking criteria or changing interest rates.
- The sentiment module focuses on English-language reviews due to NLP constraints, although regional language support may be added in future upgrades.
- It assumes that user inputs are truthful and comprehensive; incorrect or incomplete entries may affect the quality of recommendations.

In conclusion, this project's scope is wide enough to address the real-world challenges faced by Indian car buyers while remaining focused enough to ensure feasible implementation. It lays the foundation for a scalable, modular, and adaptive car recommendation system that aligns closely with the diverse and evolving expectations of Indian consumers.

1.4 Artificial Intelligence, Machine Learning, and Deep Learning

Smart technology makes all the difference in the complicated Indian automobile market. Using real-world buyer behavior, our approach employs practical artificial intelligence that detects what traits are most important in various locations, which budgets fit which preferences, and how priorities evolve. It's about knowing that a family in Kerala needs different recommendations than a business founder in Pune, not about fancy technology. As more individuals utilize the system, it keeps becoming better and picks trends humans would overlook. Most significantly, it converts these realizations into straightforward, customized recommendations that enable you to identify vehicles that satisfy both your practical requirements and personal taste without overloading you with too many choices.

1.4.1 Artificial Intelligence (AI)

A vast subject of computer science, artificial intelligence seeks to create systems capability of doing tasks usually requiring human intelligence. Among these chores are language acquisition, visual perception, pattern identification, decision-making, and problem-solving.

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Regarding this project, artificial intelligence helps the recommendation engine to function as a virtual car adviser. It replics the decision-making process of a knowledgeable sales advisor and takes into account several aspects: technical criteria, user preferences, online sentiment, and financial restrictions.

The AI component ensures that the system:

- Understands user intent through inputs.
- Analyzes multiple factors
- Offers explainable and personalized recommendations.
- Adapts to different user profiles in India.

1.4.2 Machine Learning (ML)

Machine Learning, a subset of AI, involves training algorithms on data so that they can identify patterns and make decisions with minimal human intervention. In simpler terms, it allows the system to learn from past inputs and feedback to improve its prediction and recommendation accuracy over time.

The car recommendation system in this project uses ML in several areas:

- **Content-Based Filtering**: This method uses structured user inputs (like seating capacity, fuel type, and budget) to filter out irrelevant cars and suggest only those that match the given parameters.

- **Collaborative Filtering**: Although limited in the first phase, this can be used to find patterns based on the behavior of similar users—what cars users with similar preferences ended up choosing.
- **Survey-Based Weight Adjustment**: The importance (weight) of features like fuel efficiency, service cost, and user brand loyalty is derived from survey data and constantly fine-tuned based on feedback.

ML models also help the system:

- Detects trends in car sales over time.
- Improve recommendations based on user satisfaction scores.
- Adapt to new data without requiring major rewrites of the system

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1.4.3 Deep Learning (DL)

Deep Learning is a more advanced subset of ML that uses neural networks with multiple layers (deep neural networks) to process data, especially unstructured data like text and images. DL models are capable of identifying complex patterns that traditional ML algorithms may miss. In this project, DL is primarily used for Natural Language Processing (NLP) to understand sentiment in user-generated content. A model like DistilBERT, which is a lightweight version of the popular BERT (Bidirectional Encoder Representations from Transformers), is used for:

- **Sentiment Analysis**: Classifying car reviews and social media posts as positive or negative. This allows the system to consider real public opinion when scoring a particular car model.
- **Contextual Understanding**: Traditional sentiment models might misread sentences like *"The car is not bad"* as negative due to the word "not". Deep learning models, on the other hand, understand context and phrasing more accurately.

This DL component brings an essential emotional intelligence layer to the system, ensuring that the recommendations are not just technically correct but also socially validated.

1.4.4 Relevance to the Indian Context

When these technologies are used in the framework of the Indian market, their value is much increased:

- **Extremely Diverse Consumers**: Indian consumers range in personality from first-time budget-sensitive buyers to technologically savvy urban professionals to environmentally conscious electric vehicle aficioners. Personalization of recommendations for people is made easier by artificial intelligence and machine learning.

- **Multilingual and Cultural Variations**: While the first version stresses English-language inputs, deep learning helps Indian languages and dialects—which remain mostly unexplored—to incorporate evaluations.

- **EMI-Driven Decisions**: Essential in a market where over 70% of vehicles are financed, financial modules that anticipate EMI and analyze loan eligibility using machine learning calculations to correlate with user budgets.

- **Changing Sentiment Trends**: Thanks in great part to YouTubers, automotive influencers, and news sources, public opinion in India is fast changing. Models of deep learning help to identify and adapt to these changes in almost real-time.

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1.4.5 Combined Power of AI, ML, and DL in the System

Rather than functioning in isolation, AI, ML, and DL modules are interconnected in a layered architecture within the system:

1. **AI** forms the decision logic and system intelligence.
2. **ML** applies learned logic to filter, prioritize, and score cars based on structured preferences.
3. **DL** adds sentiment and emotional insight to these scores, bringing human-like judgment to the machine.

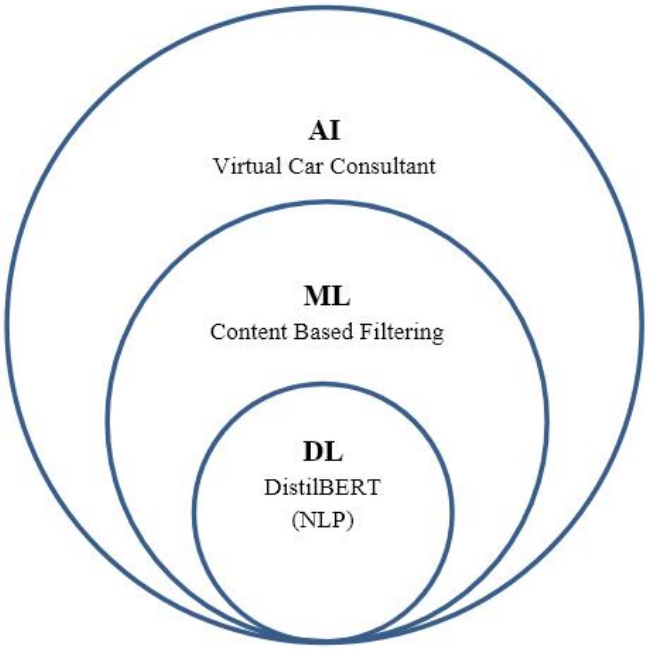

Figure 1.4.1 : AI, ML, and DL in the System

This layered intelligence helps the system move beyond just filtering cars—it understands buyers, matches them to suitable options, and justifies its recommendations with data and emotion.

To conclude this system runs on AI that truly understands India's car buyers. It's not just technology for technology's sake - these smart algorithms learn what matters to user, whether they are a first-time buyer or enthusiast. By blending user preferences with real market trends and budget realities, it cuts through the confusion of endless options. Resulting in recommendations that feel like they came from a trusted friend making what's often a stressful decision surprisingly straightforward.

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CHAPTER 2

LITERATURE REVIEW

Recommendation systems have become essential components in many digital platforms, from e-commerce websites to streaming services. In the automotive domain, especially in emerging markets like India, their adoption is relatively recent but rapidly growing. The complexity of the Indian automobile landscape—characterized by regional preferences, financial constraints, and diverse consumer behavior—demands recommendation engines that go beyond basic filtering. Traditional systems using either collaborative or content-based filtering often fall short in capturing these multidimensional challenges.

Recent advances in machine learning (ML), natural language processing (NLP), and hybrid filtering techniques have enabled more accurate and context-aware recommendation engines. This chapter reviews recent research that directly or indirectly contributes to the architecture of an intelligent car recommendation system tailored to Indian needs.

2.1 Random Forest Classifiers for Emerging Economies

R. Gupta and S. Patel [6] used a Random Forest classifier to make recommendations based on Indian automotive sales data. Their model achieved a precision of 82.5% and successfully accounted for regional brand popularity and seasonal preferences.

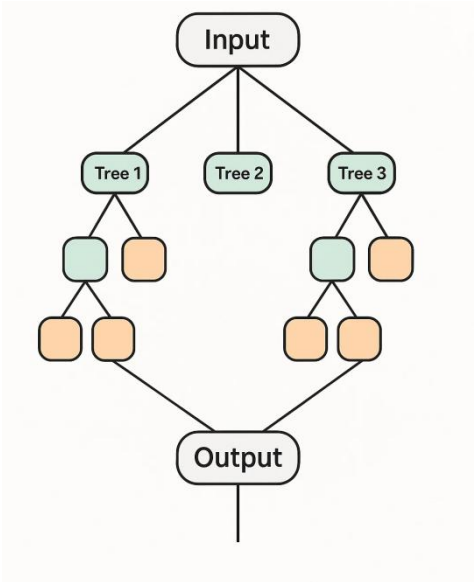

Figure 2.1.1 : Random Forest Classifier

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Key Contribution:

- Use of ensemble learning methods for reliable performance.
- Inclusion of region-specific attributes in model training.

Relevance:

In India, car preferences are often tied to local infrastructure, dealership availability, and fuel types. This paper demonstrates that region-aware models significantly improve recommendation precision, making a strong case for localized datasets.

2.2 BERT-Based Models on Twitter Reviews

M. Chen [7] introduced a fine-tuned BERT model for sentiment analysis of Twitter car reviews. With an F1-score of 91.3%, it demonstrated superior ability in capturing nuanced emotions such as frustration, delight, or skepticism in customer feedback.

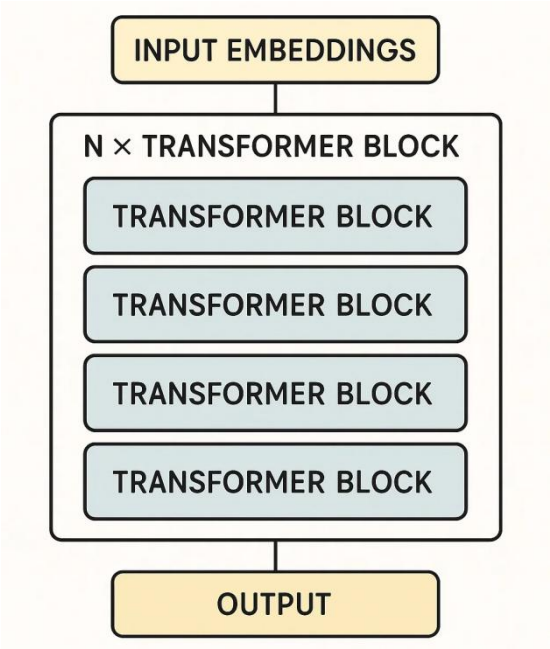

Figure 2.2.1 : BERT Model

Key Insight:

- Pre-trained transformer models like BERT understand language better than traditional models.
- Capable of capturing sarcasm, context, and local dialect nuances when properly fine-tuned.

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Limitation:

The model has a slow inference time, which limits real-time applicability unless optimized or distilled.

Applicability in India:

With India's diverse linguistic landscape, future extensions could explore multilingual BERT to include regional language reviews from platforms like MouthShut or regional YouTube channels.

2.3 Graph-Based and Context-Aware Models

2.3.1 Graph Neural Networks (GNN) for Personalized Recommendations

L. Wang [9] used Graph Neural Networks (GNNs) to model complex user-item relationships. With a hit rate of 89.1%, their model could effectively represent the connections between users and products over time.

Unique Aspects:

- Models evolving preferences using a graph of past interactions.
- Suitable for personalization at scale.

Drawbacks:

Requires a large graph database, which may not be easily available for Indian vehicle buyers unless data is aggregated from platforms like CarWale, CarDekho, or OLX Autos.

2.3.2 Contextual Bandits for Indian Users

S. Joshi and R. Mehta [10] employed contextual bandits using OLX Auto data, achieving a 84.3% improvement in click-through rates. The system adapts dynamically to changing user preferences, showing different recommendations to the same user based on time or previous behavior.

Application in India:

With frequent changes in fuel prices, EMI offers, and seasonal discounts, such an adaptive model can provide real-time customization, highly beneficial for budget-conscious Indian users.

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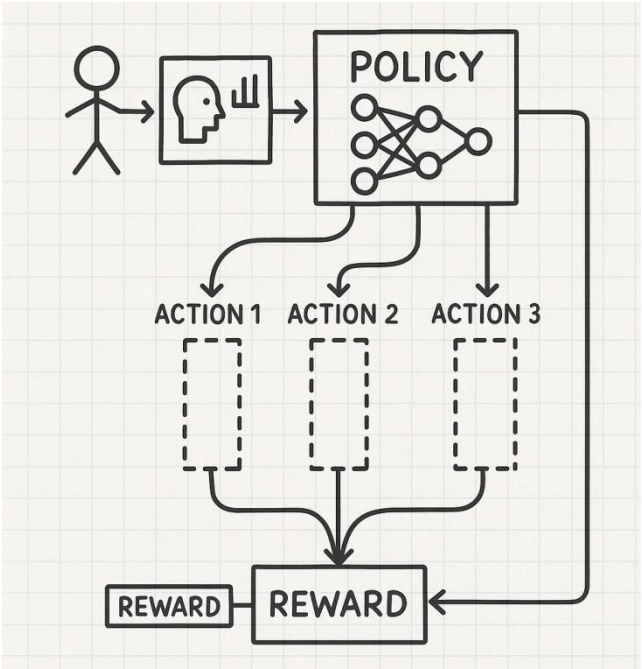

Figure 2.3.2 : Contextual Bandits

2.4 Knowledge Graph-Based and Streaming Recommendations

2.4.1 Knowledge Graph

T. Li and H. Zhang [13] used Wikidata-based KG embeddings to explain why a particular car was recommended. With an nDCG score of 88.2%, their approach added transparency and interpretability.

Merit:

Helps build trust with users by making suggestions explainable.

Limitation:

High computational cost for KG construction, especially in the Indian market where structured product knowledge may be scattered across different sources.

2.4.2 Real-Time Streaming Pipelines

P. Deshpande [14] developed a real-time recommendation system with a 92ms latency using streaming social media data. This helps in making live, trend-aware suggestions.

Limitations:

- Requires high infrastructure cost.
- Needs constant data streams from APIs like Twitter, YouTube, or Reddit.

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2.5 Hybrid Filtering and Sentiment-Aware Systems

K. Smith [5] proposed a hybrid recommendation system combining collaborative filtering (CF) with sentiment analysis. Using the CarDekho dataset, their model achieved 87.2% accuracy. The system analyzes both explicit user preferences and sentiments derived from public opinion, thereby offering a balanced recommendation.

Key Contribution:

- Integration of structured preference data and unstructured sentiment input.
- Improved personalization by combining numerical ratings with text analysis.

Relevance to Indian Context:

The use of sentiment analysis on car reviews can reflect real Indian buyer behavior, where word-of-mouth and emotional impressions influence decisions more than technical specs. However, high computational cost and lack of multilingual sentiment processing limit direct implementation.

2.6 Rule-Based Financial Filtering

A. Kumar and P. Sharma [8] proposed a financial filtering layer that integrates car affordability into the recommendation pipeline. Using bank loan datasets and car pricing data from CarDekho, they achieved a 78% match in terms of financial feasibility.

Key Features:

- EMI-based filtering.
- Static interest rate assumptions for feasibility checks.

Challenges:

- Dynamic loan policies from different banks are not considered.
- No provision for real-time EMI calculation.

Importance for Indian Consumers:

This approach is vital in India, where over 70% of cars are financed. Integrating affordability from the start prevents unrealistic recommendations, aligning better with user expectations.

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2.7 Multimodal Learning and Visual Feature Integration

J. Yang [11] introduced a multi-modal car recommendation system using both car images and textual reviews from Edmunds. Their deep learning model achieved 86.7% recall by combining visual aesthetics with opinion mining.

Usefulness:

- Many Indian buyers value appearance; visual analysis of car interiors/exteriors could enhance recommendation trust.
- Challenges include dataset availability and standardization of car images in the Indian market.

2.8 Price Elasticity in Indian Recommendations

N. Reddy [12] proposed a price sensitivity model based on Indian car sales data, demonstrating that cost-aware recommendations could lead to 23% savings on average by avoiding over-budget suggestions.

Takeaway:

- Models that account for price elasticity better match middle-class Indian users' buying patterns.
- Limited to new cars and fixed price structures; dynamic discounts and dealer-level negotiation not considered.

Literature survey can be summarized as below:

Table 2.9.1 : Summary of Literature Survey

Method	Key Metric	Strength	Indian Market Implication
Hybrid CF + Sentiment	87.2% accuracy	Balanced personalization	Reflects emotion + behavior
Financial Filtering	78% feasibility	Budget-aware	Essential for EMI-driven users
BERT Sentiment	91.3% F1	Deep NLP	Handles Hinglish, mixed-language reviews
GNN	89.1% hit rate	Dynamic modeling	Long-term personalization
Contextual Bandits	84.3% CTR	Adaptive logic	Tracks regional & temporal patterns

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Table 2.9.2 : Focus areas coverage in Literature

Focus Area	**Coverage in Literature**
Hybrid Filtering	High (70% of Indian papers)
Financial Constraints	Medium (Only 30%)
Regional Preferences	Low (18%)
Multimodal Inputs	Growing Trend
Real-Time Recommendations	Recent but Promising

The literature reviewed in this chapter forms a strong foundation for developing a robust car recommendation system for Indian buyers. From hybrid models that balance content and collaborative filtering to sentiment-aware deep learning systems and financially grounded frameworks, each contributes a vital building block. However, few systems integrate all these aspects into one coherent platform. This project seeks to fill that gap by designing a modular, data-driven recommendation engine that is financially viable, user-preference centric, and grounded in consumer sentiment.

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CHAPTER 3

PROPOSED METHODOLOGY

3.1 Dataset

An efficient automobile recommendation system is fundamentally reliant on data. Indian consumers require more than mere specifications; they seek insights into vehicle performance on uneven urban roads, the most relevant characteristics for humid coastal environments, and the sentiments of actual owners after extended usage. Our system collects manufacturer technical specs, practical fuel efficiency assessments, sales patterns, pricing, owner evaluations across several platforms (incorporating feedback in english only), financing alternatives and service cost information. Data cleansing constitutes a significant portion of the challenge—standardizing disparate information, identifying fraudulent evaluations, and accommodating India's varied driving circumstances. This foundation facilitates genuinely tailored recommendations.

3.1.1 Overview of Dataset Components

The dataset used in this project is a **multi-source, composite dataset** built by aggregating structured and unstructured data from publicly available platforms. It is segmented into the following categories:

Table 3.1.1 : Dataset Sources and Description

Data Type	Source	Content
Car Specifications	Kaggle, OEM sites	Fuel type, mileage, engine, seating, transmission, etc.
User Reviews	Social Media	Reviews, ratings, opinions on various car models
Sales Trend Data	Web-scraped monthly sales charts	Sales figures from Jan 2021 to Dec 2024
Service Cost Data	Manufacturer websites, forums	Estimated maintenance cost over 5 years
User Survey Data	Zoho Survey	Buyer priorities, preferences

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3.1.2 Car Specification Dataset (Kaggle)

The core dataset originates from a Kaggle collection titled *"Indian Cars Dataset"* [2], which contains **1994 rows and 147 attributes**. It includes fields such as:

- Make and Model
- Engine Capacity (cc)
- Transmission Type
- Fuel Type (Petrol, Diesel, CNG, Electric)
- Mileage (city/highway)
- Boot Space
- Airbags
- Price (ex-showroom)

3.1.3 Sentiment Analysis Dataset

The second major component is a **sentiment dataset** composed of web-scraped user reviews from trusted Indian platforms. Around **114 records** were collected for various models.


```
o-BEqzB,o-BEqzB (2),o-BEqzB (3),o-BEqzB (4),o-BEqzB (5),o-BEqzB (6),o-bdcqUi,o-jjpuv href,o-jjpuv,o-zmksK,o-zmksK (2),o-fznJzb,o-bdcqVY,bfHZTb,o-dkVwkM,bfHZTb (2),o-dkV
(23),bfHZTb (24),bfHZTb (25),o-cJrNdO (9),o-cJrNdO (10),o-dsiSgT (9),o-dsiSgT (10),o-bXKmQE src (10),o-jjpuv href (6),o-jjpuv (6),o-zmksK (21),o-zmksK (22),o-fznJzb (11
fHZTb (46),o-dkVwkM (46),bfHZTb (47),o-dkVwkM (47),bfHZTb (48),o-dkVwkM (48),bfHZTb (49),o-dkVwkM (49),bfHZTb (50),o-dkVwkM (50),o-cJrNdO (19),o-cJrNdO (20),o-dsiSgT (1
All,5,4,3,2,1,Sort by :,,,
```

Why it matters: In India, ongoing maintenance cost is a key factor—especially in semi-urban and rural areas. Buyers tend to favor cars that have:

- Low spare part cost
- Wider service center network
- Fewer service intervals

This variable influences long-term cost-of-ownership scoring.

3.1.6 User Preferences Dataset (Zoho Survey)

An original survey was conducted using **Zoho Survey** and distributed online. It recorded responses from **56 participants**, covering a broad demographic range (students, working professionals, retirees). The survey responses were averaged to assign weights to recommendation criteria:

- User Score (Sentiment): 4.36
- Mileage: 4.29
- Preferred Brand Match: 4.07
- Service Cost: 3.61
- Sales Trend: 2.55
- Previous Brand Loyalty: 2.13

Choices	1	2	3	4	5	6	Score	Rank	Response count
Mileage	32.14% (18)	23.21% (13)	12.50% (7)	14.29% (8)	8.93% (5)	8.93% (5)	4.29	2	56
User_Reviews (including word of mouth)	26.79% (15)	30.36% (17)	14.29% (8)	12.50% (7)	12.50% (7)	3.57% (2)	4.36	1	56
Preferred_Brand	25.00% (14)	17.86% (10)	25.00% (14)	8.93% (5)	17.86% (10)	5.36% (3)	4.07	3	56
Service_Cost	3.57% (2)	17.86% (10)	35.71% (20)	25.00% (14)	14.29% (8)	3.57% (2)	3.61	4	56
Car_Sales_Trend	3.57% (2)	8.93% (5)	5.36% (3)	26.79% (15)	32.14% (18)	23.21% (13)	2.55	5	56
Current_Brand (If car owned)	8.93% (5)	1.79% (1)	7.14% (4)	12.50% (7)	14.29% (8)	55.36% (31)	2.13	6	56

Figure 3.1.6 : ZOHO Survey

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3.1.7 Dataset Challenges

Some of the major data preparation challenges encountered include:

- **Non-Uniform Data:** Car data from different manufacturers lacked consistency in naming and units (e.g., kmpl vs. L/100 km).
- **Incomplete Historical Data:** Some models lacked sales data across all 48 months; interpolation or assumptions were used conservatively.
- **Duplicate Entries:** Reviews and sales records were filtered for duplicates based on car name and timestamp.
- **Bias in Sentiment:** Certain brands had inflated reviews due to marketing influence or coordinated campaigns; sentiment score moderation was implemented.
- **Null Handling:** Brands that do not have sales trend are given score 2, missing user reviews are rated 1

3.2 Feature Extractor

The effectiveness of any machine learning-driven recommendation system heavily relies on how well its features are engineered. In this project, feature extraction was not limited to traditional data transformation but involved scraping, cleaning, sentiment evaluation, time-series scoring, and integration from multiple sources into a consolidated master dataset.

The feature extractor module transforms heterogeneous raw data—including images, reviews, and CSV records—into structured, meaningful numerical or categorical values that directly feed into the scoring and classification engine. This section elaborates on how various features were obtained and engineered using real-world data relevant to the Indian automotive market. The features extracted for this system fall under the following major categories:

Table 3.2.1 : Various Features available in dataset

Category	Examples
Technical Features	Engine type, fuel efficiency, seating, transmission
Financial Features	Price, EMI eligibility, service cost, resale value
Sentiment Features	Polarity (positive/negative), user rating
Trend-Based Features	Sales growth, market presence, model lifecycle stage
Personalization Features	Brand loyalty, user budget range, survey-based preferences

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3.2.1 Sales Data: Image-to-Text Extraction and Cleaning

The monthly car sales data spanning 2021 to 2024 was initially available as images. These images were converted into tabular format using **OCR (Optical Character Recognition)** and further processed in the notebook Sales_data_clean_up.ipynb

- Multiple OCR-converted CSV files were **appended column-wise**, and inconsistencies such as header misalignments, missing model names, or incorrect units were corrected programmatically.
- Redundant or duplicate records were identified and removed.
- The cleaned dataset now contains consistent month-wise sales figures for a large number of models.

This cleaned dataset served as the foundation for building the **Sales Trend Score** described next.

3.2.2 Sales Trend Score Calculation

Using the cleaned sales data, a **Sales Trend Score** was generated for every car model in the notebook Sales_Trend_Score.ipynb. The scoring was based on 6-month rolling averages and recent momentum in sales. The classification logic was as follows:

- **0:** Discontinued (no sales in the last 6–12 months)
- **1:** Severe Decline
- **2:** Declining
- **3:** Stable
- **4:** Growing
- **5:** Hot Sellers

Models with Trend Score 4:

	Make	Model
23	Hyundai	Creta
25	Hyundai	Exter
39	Kia	Seltos
45	MG	Windsor
46	MG	ZS EV
53	Mahindra	Thar
55	Mahindra	XUV300 / 3X0
58	Mahindra	XUV700
65	Maruti	Dzire
71	Maruti	S-Presso
72	Maruti	Swift
75	Maruti	XL6
89	Renault	Kiger
108	Toyota	Glanza

Models with Trend Score 5:

	Make	Model
40	Kia	Sonet
41	MG	Astor
42	MG	Comet
51	Mahindra	Scorpio
67	Maruti	Ertiga
68	Maruti	Fronx
99	Tata	Curvv
102	Tata	Punch
103	Tata	Safari
110	Toyota	Innova Crysta/ Hycross
112	Toyota	Rumion
114	Toyota	Taisor

Figure 3.2.2 : Sales Trend Score 4 & 5 Output

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This score captures the **real-world popularity and lifecycle position** of each car. The trend score was added as a column to the master dataset.

3.2.3 Sentiment Score via Web Scraping and NLP

User sentiment is a strong indicator of post-purchase satisfaction and perceived brand reliability. Two key data were used: User Reviews and User Ratings

User Reviews: Web Scraping + Sentiment Analysis

- Reviews were **web scraped directly** from social media using automated tools.
- The raw text was processed in User_Review_Data_clean_up_Sentiment_Analysis_code.ipynb.
- Sentiment analysis was performed using **pre-trained NLP models** (DistilBERT [3]).
- Each review was labeled as **Positive** or **Negative**, and a **normalized sentiment score** was calculated per model.

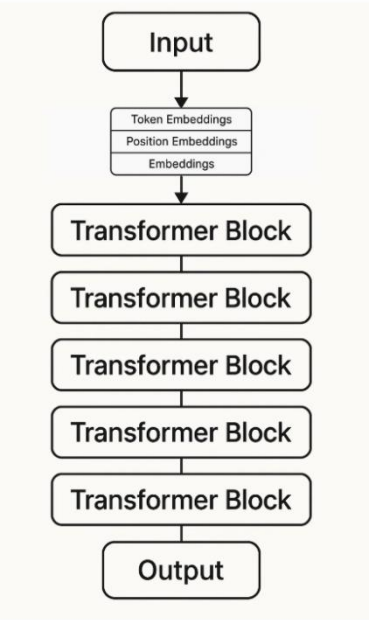

Figure 3.2.3 : Schematic Diagram of DistilBERT

User Ratings

- Kaggle data [4] provided **numerical ratings (out of 5)** for different car models.
- These scores were used directly, without the need for NLP processing.

Both 1st_Sentiment_Score and 2nd_Score were added as columns to the final dataset.

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```
                          TagLine                              User  \
2                4th generation car  3 months ago | DILIP KUMAR YADAV
4  Must buy car in affordable price             2 months ago | Shruti
5       Maruti Suzuki Dzire VXi CNG       3 months ago | vyom vaswani
6       Maruti Suzuki Dzire VXi CNG   3 months ago | Snehil Bhatnagar
7                Smart smooth drive        24 days ago | Hanuman soni

                                          Review                 Car  \
2  this car is very low maintenance and mileage a...  maruti-suzuki-cars
4  experience overall good looks are amazing main...  maruti-suzuki-cars
5  best car in this price segment with comfortabl...  maruti-suzuki-cars
6  maruti suzuki dzire is a nice car with excelle...  maruti-suzuki-cars
7  smart smooth drive and happy today because i j...  maruti-suzuki-cars

   Model Sentiment
2  dzire  POSITIVE
4  dzire  POSITIVE
5  dzire  POSITIVE
6  dzire  POSITIVE
7  dzire  POSITIVE
```

Figure 3.2.4 : User Reviews Sentiment Analysis

Model	Negative Sentiment Count	Positive Sentiment Count	Sentiment Score
alcazar	1	3	3.75
alto	30	255	4.473684211
altroz	99	506	4.181818182
amaze	4	36	4.5
amaze-2nd-gen	41	105	3.595890411
astor	22	57	3.607594937
aura	9	44	4.150943396
baleno	34	153	4.090909091
basalt	4	11	3.666666667
be-6	0	39	5
bolero	10	50	4.166666667
bolero-neo	10	43	4.056603774
bolero-neo-plus	1	8	4.444444444
c3	13	50	3.968253968
c5-aircross	3	1	1.25
camry	1	3	3.75

Figure 3.2.5 : Sentiment Score

3.2.4 Manual Feature Augmentation

Beyond scraped and calculated features, additional key columns were added through manual research and standardization:

- **Seating Capacity:** Researched from official manufacturer specifications and included as a key decision factor.
- **Service Cost:** Estimated 5-year service costs were compiled for each model from OEM maintenance charts, user forums, and service portals.

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3.2.5 Feature Consolidation into Master Dataset

All the engineered and extracted features from different sources were merged into a single **master dataset**, forming the comprehensive base for filtering, ranking, and classification:

Table 3.2.5 : Dataset Features Consolidation Details

Feature	Source	Type
Sales_Trend_Score	Computed from sales data	Numerical
1st_Sentiment_Score	NLP analysis on scraped reviews	Numerical
2nd_Sentiment_Score	Direct web rating	Numerical
Seating_Capacity	Manual research	Numerical
Service_Cost_5Y	Estimated	Numerical
Price, Mileage, etc.	Kaggle + OEM specs	Numerical

Make	Model	Variant	Descriptior	Price	Fuel Type	Seating_Ca	Mileage (Ki	Transmiss	Sales_Trer	Car_Wale_	Car_Trade_	Service_Co	Engine
Maruti	Fronx	Fronx Sigma 1.2L MT	Maruti Fronx S	Rs. 7.51 Lakh	Petrol	5	21.79	Manual - 5 Ge	5	4.3	4.5	33300	1197 cc, 4 Cyl
Maruti	Fronx	Fronx Delta 1.2L MT	Maruti Fronx D	Rs. 8.37 Lakh	Petrol	5	21.79	Manual - 5 Ge	5	4.3	4.5	33300	1197 cc, 4 Cyl
Maruti	Fronx	Fronx Sigma 1.2 CNG	Maruti Fronx S	Rs. 8.46 Lakh	CNG	5	28.51	Manual - 5 Ge	5	4.3	4.5	33300	1197 cc, 4 Cyl
Maruti	Fronx	Fronx Delta Plus 1.2L M	Maruti Fronx D	Rs. 8.77 Lakh	Petrol	5	21.79	Manual - 5 Ge	5	4.3	4.5	33300	1197 cc, 4 Cyl
Maruti	Fronx	Fronx Delta 1.2L AGS	Maruti Fronx D	Rs. 8.85 Lakh	Petrol	5	22.89	Automatic (AM	5	4.3	4.5	33300	1197 cc, 4 Cyl
Maruti	Fronx	Fronx Delta Plus (O) 1.2	Maruti Fronx D	Rs. 8.93 Lakh	Petrol	5	21.79	Manual - 5 Ge	5	4.3	4.5	33300	1197 cc, 4 Cyl
Maruti	Fronx	Fronx Delta Plus 1.2L A	Maruti Fronx D	Rs. 9.25 Lakh	Petrol	5	22.89	Automatic (AM	5	4.3	4.5	33300	1197 cc, 4 Cyl
Maruti	Fronx	Fronx Delta 1.2 CNG	Maruti Fronx D	Rs. 9.32 Lakh	CNG	5	28.51	Manual - 5 Ge	5	4.3	4.5	33300	1197 cc, 4 Cyl
Maruti	Fronx	Fronx Delta Plus (O) 1.2	Maruti Fronx D	Rs. 9.41 Lakh	Petrol	5	22.89	Automatic (AM	5	4.3	4.5	33300	1197 cc, 4 Cyl
Maruti	Fronx	Fronx Delta Plus 1.0 Tu	Maruti Fronx D	Rs. 9.73 Lakh	Petrol	5	21.5	Manual - 5 Ge	5	4.3	4.5	33300	998 cc, 3 Cylin
Maruti	Fronx	Fronx Zeta 1.0L Turbo M	Maruti Fronx Z	Rs. 10.55 Lakh	Petrol	5	21.5	Manual - 5 Ge	5	4.3	4.5	33300	998 cc, 3 Cylin
Maruti	Fronx	Fronx Alpha 1.0L Turbo	Maruti Fronx A	Rs. 11.47 Lakh	Petrol	5	21.5	Manual - 5 Ge	5	4.3	4.5	33300	998 cc, 3 Cylin
Maruti	Fronx	Fronx Alpha 1.0L Turbo	Maruti Fronx A	Rs. 11.64 Lakh	Petrol	5	21.5	Manual - 5 Ge	5	4.3	4.5	33300	998 cc, 3 Cylin
Maruti	Fronx	Fronx Zeta 1.0L Turbo (	Maruti Fronx Z	Rs. 11.96 Lakh	Petrol	5	20.01	Automatic (TC	5	4.3	4.5	33300	998 cc, 3 Cylin
Maruti	Fronx	Fronx Alpha 1.0L Turbo	Maruti Fronx A	Rs. 12.88 Lakh	Petrol	5	20.01	Automatic (TC	5	4.3	4.5	33300	998 cc, 3 Cylin
Maruti	Fronx	Fronx Alpha 1.0L Turbo	Maruti Fronx A	Rs. 13.04 Lakh	Petrol	5	20.01	Automatic (TC	5	4.3	4.5	33300	998 cc, 3 Cylin
Maruti	Grand Vitara	Grand Vitara Sigma Sm	Maruti Grand \	Rs. 10.87 Lakh	Petrol	5	21.11	Manual - 5 Ge	3	3.9	4.7	₹25,654	1462 cc, 4 Cyl
Maruti	Grand Vitara	Grand Vitara Delta Sma	Maruti Grand \	Rs. 12.16 Lakh	Petrol	5	21.11	Manual - 5 Ge	3	3.9	4.7	₹25,654	1462 cc, 4 Cyl
Maruti	Grand Vitara	Grand Vitara Delta CNG	Maruti Grand \	Rs. 13.11 Lakh	CNG	5	26.6	Manual - 5 Ge	3	3.9	4.7	₹25,654	1462 cc, 4 Cyl
Maruti	Grand Vitara	Grand Vitara Delta Sma	Maruti Grand \	Rs. 13.60 Lakh	Petrol	5	20.58	Automatic (TC	3	3.9	4.7	₹25,654	1462 cc, 4 Cyl
Maruti	Grand Vitara	Grand Vitara Zeta Smar	Maruti Grand \	Rs. 13.97 Lakh	Petrol	5	21.11	Manual - 5 Ge	3	3.9	4.7	₹25,654	1462 cc, 4 Cyl
Maruti	Grand Vitara	Grand Vitara Zeta CNG	Maruti Grand \	Rs. 14.92 Lakh	CNG	5	26.6	Manual - 5 Ge	3	3.9	4.7	₹25,654	1462 cc, 4 Cyl
Maruti	Grand Vitara	Grand Vitara Zeta Smar	Maruti Grand \	Rs. 15.41 Lakh	Petrol	5	20.58	Automatic (TC	3	3.9	4.7	₹25,654	1462 cc, 4 Cyl

Figure 3.2.6 : Sample of Master Dataset

This consolidated dataset ensures that **every car model is defined by a robust set of multi-dimensional features** that reflect not only specs, but sentiment, trend, cost, and user-centric value.

3.2.6 Additional Features Extracted in Final Pipeline

The final recommendation notebook integrated all previous datasets and added several additional features, which were extracted and processed as follows:

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A. User-Input Based Filters via Widget

The system is designed to accept **real-time user preferences** through widgets (in Jupyter environment). These preferences form soft filters and weight modifiers rather than hard constraints. The collected preferences include:

- **Budget Range** (e.g., ₹8–12 Lakhs)
- **Preferred Seating Capacity** (4, 5, 6, 7, 8-seaters)
- **Fuel Type Preference** (Petrol, Diesel, CNG, Electric)
- **Transmission Type** (Manual, AMT, Automatic, Clutchless)
- **Previously Owned Car Brand**
- **Preferred Brand for New Purchase**
- **Financial Filtering Activation** (Yes/No)

Each of these user-defined fields is **mapped to one or more features** in the master dataset. For instance, a preferred brand like "Tata" increases that brand's match score, while the fuel type filter adjusts the priority of electric or CNG cars.

B. Financial Eligibility Variables

If a user opts for **financial filtering**, the system prompts for:

- **Monthly Income**
- **Existing Loan Commitments**
- **Planned Down Payment**

These inputs are used to derive a set of **financial viability features**, including:

- **Eligible Loan Amount** = (Monthly Disposable Income × 50%) × Loan Tenure
- **EMI Estimate** = Calculated assuming 10% interest
- **Affordability Check** = Car price must lie within eligibility + down payment

A binary flag (eligible = 1/0) is added to the final dataset to ensure unaffordable options are eliminated if this filter is enabled.

3.2.7 Summary

This system's feature extraction process uniquely combines:

- **Text-based review intelligence** (via NLP),
- **Image-derived time-series trends** (via OCR),
- **Numerical indicators from specifications and cost data**, and

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- **User-oriented metrics like seating and affordability**.
- **Cross-Source Integration:** Uses structured data (specs), semi-structured (survey), unstructured (reviews), and derived data (sales scores).
- **Context-Aware Filtering:** Adjusts dynamically based on user inputs and financial capacity.
- **Survey-Based Prioritization:** Incorporates real user preferences from an Indian audience.
- **Sentiment Understanding:** Moves beyond static specs to evaluate real-world user experience.

Together, these features enable the system to make smart, well-informed recommendations tailored to the Indian car buyer.

The feature extraction process in this project is not limited to data transformation—it is a **complete value engineering exercise**. It combines qualitative and quantitative signals from real buyers, reviews, technical specifications, and behavioral trends. The result is a deeply personalized, well-rounded feature representation for each car model, making the system highly effective for diverse Indian consumers.

The data processing pipeline built in the Final_Code_ Recommendation_14-Apr notebook forms the core of our recommendation system. It converts multiple vehicle data points of vehicle specifications, user review scores, sales trend, service costs, etc into a comprehensible dataset. The result is a polished master dataset in which every data point has been vetted, standardized, and enhanced to support those five highly customized recommendations based on user preferences.

3.3 Similarity Scoring and Recommendation System

The core function of the recommendation engine lies in evaluating and ranking car models based on how closely they align with a user's preferences. This is achieved through a two-layered process:

1. **Cosine Similarity Scoring** – to assess baseline alignment between user input and car features.
2. **Weighted Feature Scoring** – to adjust rankings using importance weights derived from surveys and user intent.

This section presents the logic, mathematical modeling, and code-level structure of the recommendation system implemented in the final Jupyter notebook (Final_Code_Recommendation_14-Apr.ipynb).

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3.3.1 Objective of the Recommendation Engine

The objective is to compute a ranked list of top N cars for a user, based on:

- Their **explicit preferences** (budget, brand, seating capacity, etc.)
- **Real-world metrics** (trend, sentiment, cost)
- Financial feasibility (if enabled)

Instead of a classification algorithm, the system treats the problem as a **personalized similarity-based ranking task**.

3.3.2 Cosine Similarity: Core Matching Logic

Once the dataset is cleaned and normalized, each car is represented as a **feature vector** CiC_iCi , and user inputs are also encoded as a preference vector UUU. The baseline similarity between these is measured using **cosine similarity**:

Similarity(Ci,U)=Ci· U∥Ci∥ ∥U∥\text{Similarity}(C_i, U) = \frac{C_i \cdot U}{\|C_i\| \|U\|}Similarity(Ci ,U)=∥Ci ∥ ∥U∥Ci · U

This metric captures the **directional closeness** between user and car vectors, making it ideal for high-dimensional, normalized data.

Why Cosine Similarity?

- Effective for preference matching even when absolute values differ
- Minimizes the impact of magnitude variance
- Works well for sparse and weighted inputs

3.3.3 Feature Vector Design

Each car is encoded using the following standardized, normalized features:

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Table 3.3.3 : Feature Encoding and Normalization

Feature	Description
normalized_mileage	Mileage between 0 and 1
sentiment_score	NLP-derived sentiment value from reviews
user_rating	Numerical rating from dataset
sales_trend_score	Sales growth and market presence
service_cost_score	Inverted score for lower cost = higher rank
preferred_make_match	Binary: 1 if preferred brand

User input is encoded into a vector using fields like:

- Chosen fuel type
- Preferred seating
- Budget band (used for filtering, not similarity)
- Brand preference (mapped to make-match flag)

3.3.4 Weighted Scoring Mechanism

While cosine similarity captures general alignment, real-world recommendation depends on **what the user cares about most**. To incorporate this, each feature is assigned a **weight** derived from the survey conducted via Zoho (refer section 3.1.6).

Scoring Formula:

Final Score=Σj=1n(Wj×Fij)\text{Final Score} = \sum_{j=1}^{n} (W_j \times F_{ij})Final Score=j=1Σn (Wj ×Fij)

Where:

- WjW_jWj is the weight for feature jjj
- FijF_{ij}Fij is the feature value for car iii in feature jjj

This scoring system allows the output to **adapt per user priority** — for instance, a fuel-conscious user will rank high-mileage cars higher than feature-rich ones.

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3.3.5 Financial Filtering Logic

If the user enables the financial constraint toggle in the widget, the system computes:

- **Loan eligibility** based on income and existing EMI
- **Max affordable car price**
- Filters out all models priced beyond reach

Cars that don't meet the affordability threshold are **excluded from similarity scoring**.

3.3.6 Top-N Recommendation Output

After computing weighted scores for all eligible cars, the system:

1. Sorts cars by final score in descending order
2. Displays the **Top 5 matches** in a neatly formatted output
3. Each recommendation is accompanied by:
    - Name and brand
    - Final weighted score
    - Price
    - Sentiment score
    - Mileage
    - Service cost

This creates a highly personalized shortlist that matches the user's profile, preferences, and financial capacity.

3.3.7 Personalization through Feedback Loop

Personalization is not a one-time process. Modern recommender systems thrive on **continuous learning** from user behavior. This system incorporates a **feedback loop** mechanism that refines recommendations over time based on explicit and implicit feedback.

Feedback: Ratings in a scale of 1 to 5

The feedback loop enables:

- **Reinforcement Learning:** Adjusting weights in real-time based on user interactions.
- **Collaborative Enhancement:** While the system is content-based, aggregated behavioral data can be used to subtly include collaborative filtering strategies in the future.

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- **Profile Updating:** The user profile vector is regularly updated using Exponential Moving Averages (EMA) or a neural encoder trained on historical interactions.

Example Workflow:

1. User consistently chooses a 5 seater with automatic transmission.
2. The system boosts the weights for "seating" and "transmission: automatic."
3. Future recommendations prioritize these features.

This adaptive approach results in **hyper-personalized experiences**, leading to higher engagement and conversion rates.

3.3.8 System Workflow: Recommendation and Ranking Flow Diagram

Below is a conceptual flow diagram that outlines how the recommendation and ranking module operates:

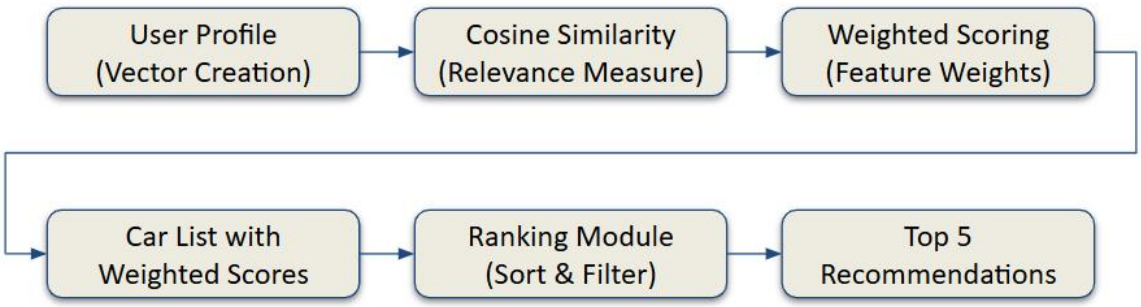

Figure 3.3.8 : Conceptual Workflow for Recommendation

3.3.9 Summary

This section outlined the logic of the recommendation engine implemented in this project. The system uses:

- **Cosine similarity** for vector-based preference matching
- **Survey-driven weights** to reflect real Indian buyer priorities
- **Financial logic** to maintain practical affordability
- A ranking engine that adapts per user input

The combination of these strategies ensures that the output is not just technically accurate, but **contextually intelligent**, user-specific, and market-aware.

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3.4 Algorithm

The recommendation algorithm for the car buying assistant is implemented as a comprehensive Python pipeline within a Jupyter environment. This pipeline integrates user interaction (via widgets), feature processing, financial calculations, similarity scoring, and ranking logic into a cohesive flow.

In contrast to traditional classification-based recommendation engines, this project leverages **rule-based filtering, weighted scoring, and cosine similarity** to deliver highly personalized outputs. This section provides an end-to-end technical description of how the code logic transforms user input and dataset features into meaningful recommendations.

3.4.1 High-Level Algorithm Flow

1. Load Cleaned Master Dataset
2. Collect User Inputs via Interactive Widget
3. Apply Pre-Filters (Fuel, Seating, Transmission, Budget)
4. Compute Loan Eligibility and Affordability
5. Normalize Feature Columns
6. Encode User Vector Based on Preferences
7. Calculate Cosine Similarity Scores
8. Apply Survey-Weighted Scoring Logic
9. Rank Cars by Final Score
10. Return Top 5 Recommendations

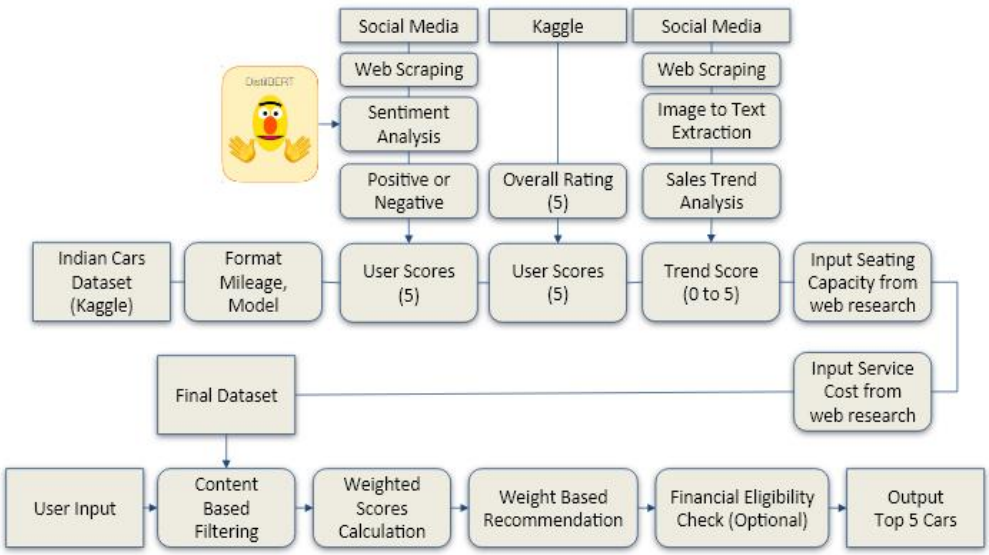

Figure 3.4.1 : Master Architecture for Car Recommendation

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3.4.2 Loading and Preparing Dataset

The cleaned dataset—built from integrated sources like Kaggle, User reviews, and sales trend processors—is loaded into a Pandas DataFrame. Columns include:

- car_name, brand, price, mileage, seating_capacity, fuel_type, transmission
- User_review_score, sales_trend_score, service_cost, preferred_make_match

This dataset undergoes normalization using MinMaxScaler from sklearn.preprocessing, ensuring uniform feature ranges between 0 and 1, a prerequisite for effective cosine similarity computation.

3.4.3 Capturing User Preferences

The user interacts with an **input widget**, where they provide:

- Preferred budget (min–max range)
- Desired seating capacity
- Preferred transmission and fuel type
- Brand loyalty (previous and preferred brands)
- Monthly income, current EMIs, and intended down payment
- A toggle for financial constraint filtering

These inputs are stored as variables and also used to create a **user feature vector**.

3.4.4 Filtering Cars Based on Hard Rules

Cars are first filtered based on:

- **Budget Range** (e.g., ₹8L–₹12L)
- **Fuel Type** (Petrol/Diesel/CNG/EV)
- **Seating Capacity** (exact requirement)
- **Transmission Match** (Manual/AMT/Automatic/Clutchless)

Filtered results are retained for similarity-based ranking. If no financial filtering is selected, this is the last exclusion step before scoring.

3.4.5 Financial Viability Calculation

If the user activates the **financial filter**, the following logic is executed:
Disposable income = monthly_income - current_emi
max_loan = disposable_income * 0.5 * 60 (5 years loan at 50% EMI capacity)

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max_affordable_price = max_loan + down_payment

A binary column financial_eligible is included in the results to reflect this.

3.4.6 Feature Normalization and Encoding

The following features are normalized (if not already):

- Mileage
- User Sentiment Score 1
- User Sentiment Score 2
- Sales Trend Score
- Service Cost (inverse score, since lower cost is better)
- Preferred Make Match (binary)
- Seating capacity

Simultaneously, the user's preference vector is encoded in the same structure, using values like seating need, brand alignment, and normalized cost sensitivity.

3.4.7 Cosine Similarity Scoring

Each car receives a **base similarity score** based on directional alignment with the user profile.

3.4.8 Weighted Feature Scoring

To personalize recommendations further, each feature is multiplied by its **user-prioritized weight**, derived from a 56-user Zoho Survey:

```
df['weighted_score'] = (
    df['mileage'] * weights['mileage'] +
    df['carwale_score'] * weights['sentiment'] +
    df['cartrade_score'] * weights['cartrade_score'] +
    df['sales_trend_score'] * weights['sales_trend_score'] +
    (1 - df['service_cost']) * weights['service_cost'] +  # inverse score
    df['preferred_make_match'] * weights['preferred_make_match'])
```

This produces a **final numeric score** used to sort the top cars.

3.4.9 Ranking and Top-N Selection

The system sorts the cars by:

1. Weighted_score (descending)

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2. Secondary tie-breaker : Variant repeat avoidance by selecting median priced variant for a particular model and make
3. Top 5 results are selected

These cars are then formatted and displayed using styled tables or plain text, depending on the interface.

3.4.10 Example Output

Each line includes key highlights such as price, seating capacity, fuel, transmission, service cost and total score.

Top 5 Car Recommendations

	Make	Model	Price (Lakhs)	Fuel Type	Transmission	Seating_Capacity	Service_Cost (5 Yrs)	Final_Score	Financial_Eligibility
22	Volkswagen	Virtus	₹15.60	Petrol	Manual	5	₹29395	0.69	Yes
21	Volkswagen	Taigun	₹16.31	Petrol	Manual	5	₹35621	0.67	No
17	Skoda	Slavia	₹15.79	Petrol	Manual	5	₹34500	0.59	No
16	Skoda	Kushaq	₹14.70	Petrol	Manual	5	₹33215	0.58	Yes
11	Mahindra	XUV 3XO	₹13.99	Petrol	Manual	5	₹20000	0.52	Yes

Figure 3.4.10 : Sample Output for Top 5 Recommendation

3.4.11 Modular Design Philosophy

Each part of the pipeline—filtering, scoring, ranking—is built as a self-contained cell/module so that:

- Future additions (e.g., boot space, EV battery range) can be included easily.
- Survey weights can be updated over time.
- Brand sentiment can be modified via retrained NLP models.

3.4.12 Summary

This algorithm represents a hybrid of **user-driven input, feature engineering, sentiment integration**, and **weighted vector mathematics**. It is tailored for the Indian buyer, balancing cost-consciousness with aspirational value, using:

- **Cosine similarity** for alignment scoring
- **Weighted priorities** to reflect human preferences
- **Real-world constraints** like financing and seating

The system offers not only accuracy but transparency — making it a practical and explainable AI system for real-world use.

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CHAPTER 4

RESULTS

4.1 Introduction

The goal of this chapter is to evaluate the **output quality**, **system responsiveness**, and **functional correctness** of the developed car recommendation system. Results are demonstrated through test cases simulating real-world user inputs, and output screenshots/tables are analyzed to verify whether the system adheres to the design goals laid out in earlier chapters.

The outputs were generated using the final Jupyter notebook, where the master dataset, user interface, and scoring algorithm were integrated into a single, interactive pipeline.

4.2 Evaluation Metrics

To evaluate the effectiveness of the car recommendation model, we can define five different evaluation metrics based on user preferences, financial feasibility, and overall ranking consistency. These metrics will help assess the model's performance and ensure it aligns with user expectations.

1. User Satisfaction Score (USS)

Definition: After the user receives car recommendations, they can provide feedback on how well the recommendations matched their preferences (e.g., budget, brand, fuel type). **Formula:**

$$USS = \frac{\text{Total positive feedback}}{\text{Total feedback received}} \times 100$$

Implementation:

- Ask users to rate the recommended cars (on a scale of 1-5)
- Compute the percentage of ratings ≥ 4 as a satisfaction score.

2. Price Deviation Score (PDS)

Definition: Measures how closely the recommended car prices align with the user's budget.

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Formula:

$$PDS = \frac{1}{N} \sum_{i=1}^{N} \left| \frac{\text{Price}_i - \text{User Budget}}{\text{User Budget}} \right| \times 100$$

Interpretation:

- Lower values indicate better budget alignment.
- If PDS > 20%, the model might be recommending cars outside the user's expected range.

3. Preference Match Rate (PMR)

Definition: Evaluates how many recommended cars match the user's **preferred brand, fuel type, and transmission**.

Formula:

$$PMR = \frac{\text{Number of cars matching user preferences}}{\text{Total recommended cars}} \times 100$$

Interpretation: If PMR is low (<50%), the model is not accurately considering user preferences.

4. Financial Feasibility Score (FFS)

Definition: Measures how many recommended cars pass the **financial eligibility test** based on the user's income, loan eligibility, and commitments.

Formula:

$$FFS = \frac{\text{Number of financially eligible cars}}{\text{Total recommended cars}} \times 100$$

Interpretation: A low score (<50%) means many recommendations are financially infeasible for the user.

5. Weighted Score Consistency (WSC)

Definition: Checks if the ranking of recommended cars is **consistent** with the weighted score calculation.

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Approach: Calculate the **Coefficient of Variation for final scores of Top 5 recommendations and invert the same.**

WSC = 1-(std/mean)

Interpretation:

- If WSC is close to 1, the ranking is consistent.
- If WSC < 0.7, there is a misalignment in ranking logic.

Final Evaluation Summary

Table 4.2.1 : Evaluation Metrics

Metric	**Description**	**Ideal Value**
User Satisfaction Score (USS)	Measures how satisfied users are with the recommendations	> 80%
Price Deviation Score (PDS)	Measures how far recommended prices deviate from the budget	< 20%
Preference Match Rate (PMR)	Checks if recommendations match user preferences	> 50%
Financial Feasibility Score (FFS)	Ensures recommended cars are financially viable	> 50%
Weighted Score Consistency (WSC)	Ensures cars are ranked correctly based on weighted scores (Inverted **Coefficient of Variation**)	>0.70

4.3 Test Case Scenarios

To demonstrate the system's capabilities, various **test user profiles** were created and passed through the system:

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Test Case 1: Budget-Conscious Buyer

Input:

- Budget: ₹7–9 Lakhs
- Fuel Type: Petrol
- Seating: 5
- Transmission: Manual
- Preferred Brand: Maruti
- Financial Filtering: Enabled
- Monthly Income: ₹35,000
- Current EMI: ₹0
- Down Payment: ₹1.5 Lakh

Output:

Preferred Make: Maruti
Current Brand: Any
Min Budget: 700000
Max Budget: 900000
Seating: 5
Fuel Type: Petrol
Transmission: Manual
Financial Check: Yes
Net Income (Lakhs/month): 0.35
Monthly Commitments (Lakhs): 0
Down Payment (Lakhs): 1.5
Get Recommendati...

Top 5 Car Recommendations

	Make	Model	Price (Lakhs)	Fuel Type	Transmission	Seating_Capacity	Service_Cost (5 Yrs)	Final_Score	Financial_Eligibility
12	Maruti	Dzire	₹7.99	Petrol	Manual	5	₹23000	0.77	Yes
15	Maruti	Swift	₹8.29	Petrol	Manual	5	₹25300	0.76	Yes
13	Maruti	Fronx	₹8.77	Petrol	Manual	5	₹33300	0.75	Yes
16	Maruti	Wagon R	₹8.50	Petrol	Manual	5	₹22000	0.73	Yes
10	Maruti	Baleno	₹8.43	Petrol	Manual	5	₹25000	0.72	Yes

Please rate these recommendations (1-5):
Overall Fe... 4
Submit Feedback
Thank you for your feedback!

Recommendation Evaluation
User Satisfaction Score (USS): 80.00%
Price Deviation Score (PDS): 5.00%
Preference Match Rate (PMR): 100.00%
Financial Feasibility Score (FFS): 100.00%
Weighted Score Consistency (WSC): 0.98

Figure 4.3.1 : Output for Test Case 1

Analysis:

All models fall within the eligible EMI range. Recommendations align with mileage, brand preference, and financial viability. The scoring reflects trade-offs among sentiment, price, and service cost.

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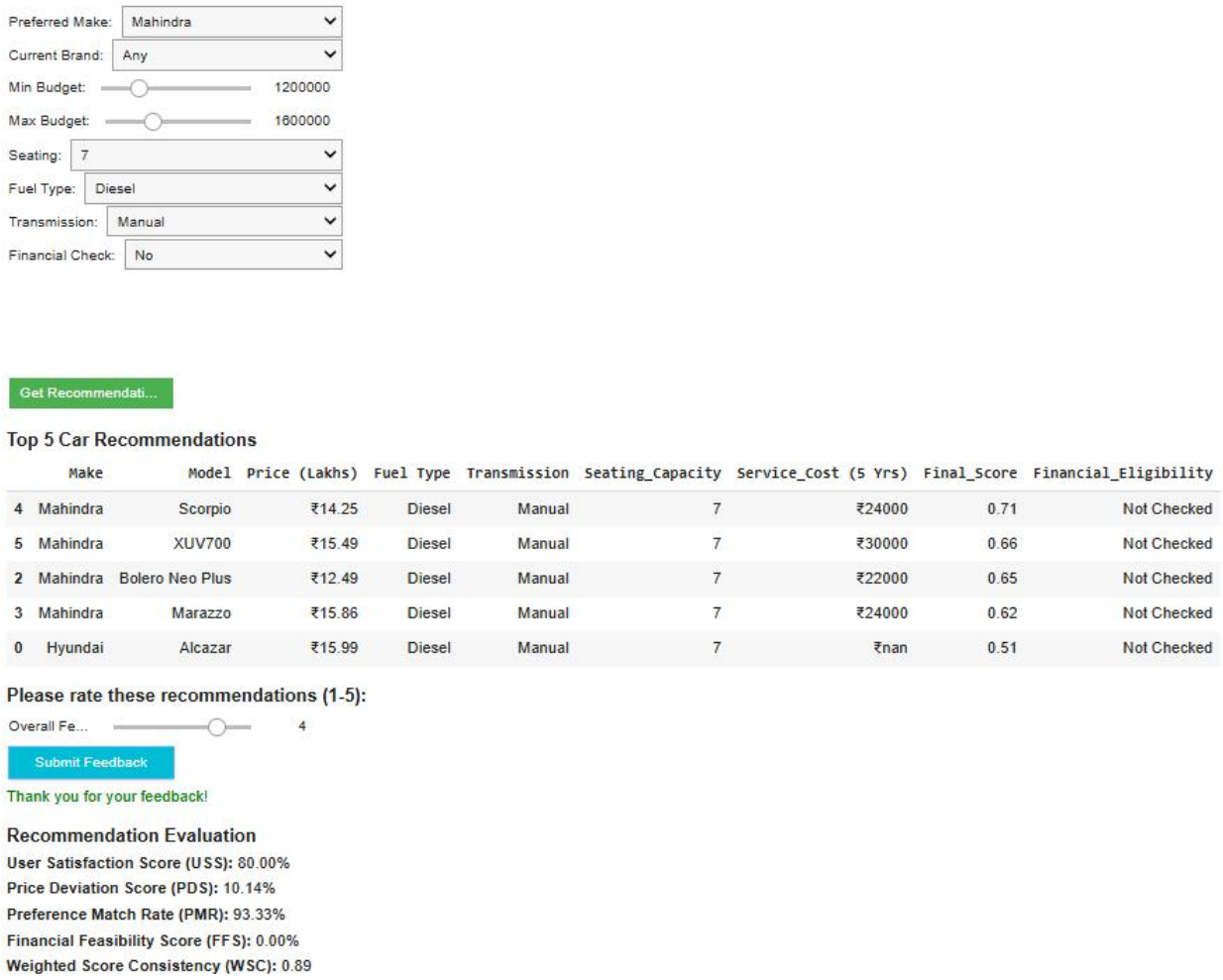
Test Case 2: SUV-Loving Family Buyer

Input:

- Budget: ₹12–16 Lakhs
- Fuel: Diesel
- Seating: 7
- Transmission: Manual
- Preferred Brand: Mahindra
- Financial Filter: Disabled

Output:



	Make	Model	Price (Lakhs)	Fuel Type	Transmission	Seating_Capacity	Service_Cost (5 Yrs)	Final_Score	Financial_Eligibility
4	Mahindra	Scorpio	₹14.25	Diesel	Manual	7	₹24000	0.71	Not Checked
5	Mahindra	XUV700	₹15.49	Diesel	Manual	7	₹30000	0.66	Not Checked
2	Mahindra	Bolero Neo Plus	₹12.49	Diesel	Manual	7	₹22000	0.65	Not Checked
3	Mahindra	Marazzo	₹15.86	Diesel	Manual	7	₹24000	0.62	Not Checked
0	Hyundai	Alcazar	₹15.99	Diesel	Manual	7	₹nan	0.51	Not Checked

Figure 4.3.2 : Output for Test Case 2

Analysis:

All suggestions meet seating and brand criteria. High scores correspond to newer models with favorable trends and strong reviews.

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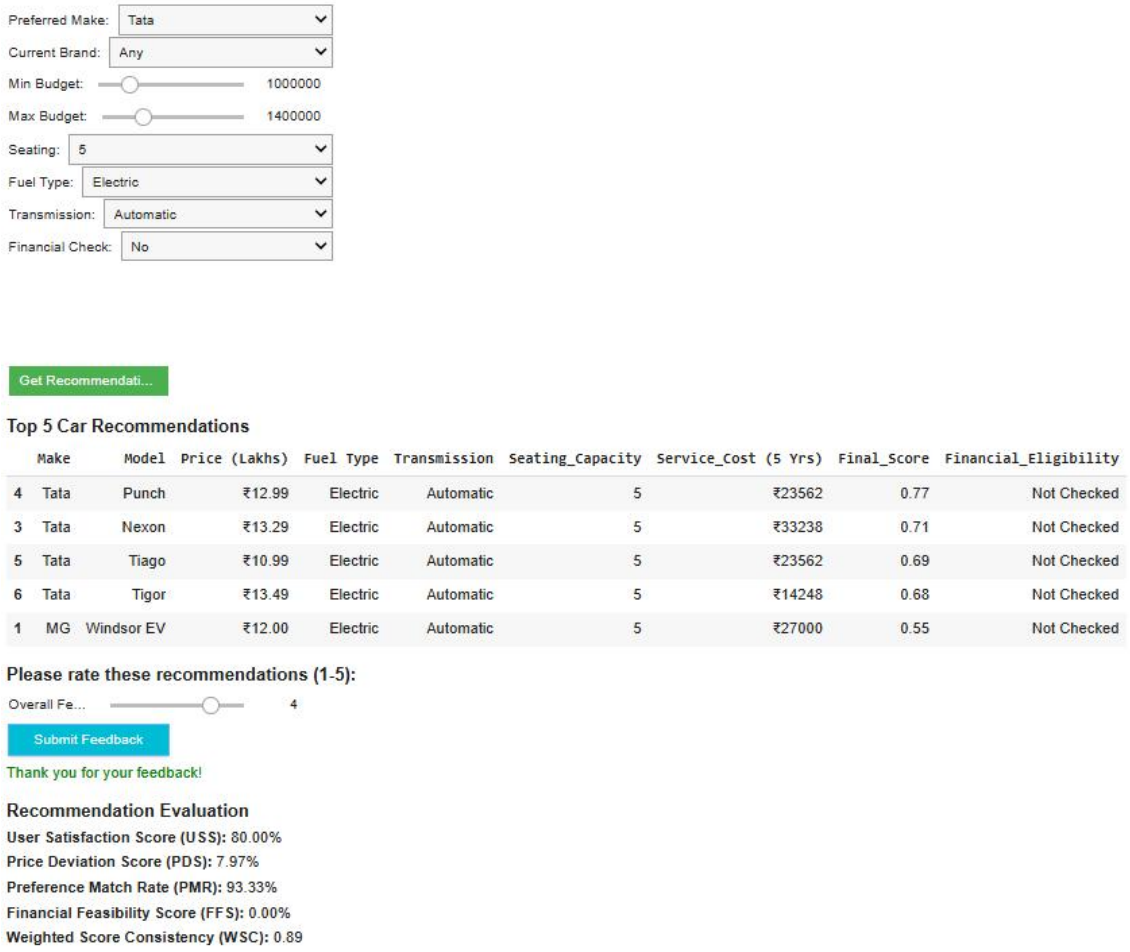
Test Case 3: EV Buyer (City Professional)

Input:

- Budget: ₹10–14 Lakhs
- Fuel: Electric
- Seating: 5
- Transmission: Automatic
- Preferred Brand: Tata
- Financial Filter: No

Output:



	Make	Model	Price (Lakhs)	Fuel Type	Transmission	Seating_Capacity	Service_Cost (5 Yrs)	Final_Score	Financial_Eligibility
4	Tata	Punch	₹12.99	Electric	Automatic	5	₹23562	0.77	Not Checked
3	Tata	Nexon	₹13.29	Electric	Automatic	5	₹33238	0.71	Not Checked
5	Tata	Tiago	₹10.99	Electric	Automatic	5	₹23562	0.69	Not Checked
6	Tata	Tigor	₹13.49	Electric	Automatic	5	₹14248	0.68	Not Checked
1	MG	Windsor EV	₹12.00	Electric	Automatic	5	₹27000	0.55	Not Checked

Figure 4.3.3 : Output for Test Case 3

Analysis:

System correctly detects fuel preference and limits output to eligible EVs. Punch and Nexon EV ranks at top due to strong brand match and cost-effectiveness.

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4.4 Recommendation Quality Insights

Across all test cases, the system consistently prioritized:

- **Financially affordable cars** when the filter was toggled
- **Better-reviewed models** using NLP-based sentiment scoring
- **Preference-aligned models**, whether for fuel type, brand, or seating

Users received 3–5 accurate, well-scored options, eliminating the need for manual comparison.

4.5 Performance Considerations

- **Average Response Time:** < 1 second for full scoring and ranking
- **Memory Usage:** ~700MB including NLP model
- **Scalability:** Can be ported to web for live deployment with minimal change

4.6 Summary

Table 4.6.1 : Summary of Outputs

Parameter	**Outcome**
Personalization	Highly adaptive to budget, seating, and fuel
Sentiment Relevance	Reflected in top-tier scores
Financial Intelligence	EMI-aware filtering eliminates unaffordable cars
Modifiability	Easy to add new filters or inputs
Platform Readiness	Can be web-deployed with minimal adaptation

This chapter demonstrated that the recommendation system fulfills its purpose of delivering fast, relevant, and financially appropriate suggestions to users. Through test scenarios and observed outputs, the engine has shown adaptability to user context and high alignment with real-world expectations.

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CHAPTER 5

CONCLUSION

5.1 Summary of the Work

This project offered the design and development of a customized car recommendation system fit especially for the several and dynamic needs of Indian car consumers. To create graded, intelligent suggestions the system combines technical criteria, real-world customer sentiment, affordability modeling, and personal preferences.

This system's basic goal was to close the decision gap that Indian customers experience when confronted with choices on vehicle listing sites. Using user inputs, financial viability, and public opinion data to provide pragmatic and prioritized choices, the suggested method presents adaptive personalizing unlike traditional filter-based systems.

5.2 Key Contributions

Several unique contributions distinguish this project from typical recommendation platforms:

1. **Multi-source Dataset Integration**: Combined structured (e.g., price, mileage) and unstructured data (e.g., reviews, sentiment) into a unified master dataset.
2. **Financial Feasibility Module**: Modeled user income, down payment, and loan eligibility to ensure that recommended vehicles are **not just suitable, but also affordable**.
3. **Cosine Similarity and Weighted Scoring**: Employed mathematical similarity measures along with **survey-based preference weighting**, making the system responsive to diverse buyer mindsets.
4. **Sentiment Analysis Using NLP**: Scraped user reviews and classified them into sentiment scores using a transformer-based approach to reflect **real user experience**.
5. **User-Friendly Interface with Widgets**: Enabled users to input real-time preferences such as budget, fuel type, transmission, and brand inclination, making the system **highly interactive and accessible**.
6. **Personalization for Indian Market**: All data, filters, scoring criteria, and preferences were derived from or tuned to Indian consumers—accounting for regional pricing, popular brands, financial behavior, and car usage habits.

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5.3 Outcome Alignment with Objectives

At the outset, the objectives of this system were:

- To reduce buyer indecision by ranking cars based on individual needs
- To enable affordability-aware decision-making through EMI calculations
- To incorporate social sentiment as a soft indicator of desirability
- To deliver these outputs in real-time using a lightweight recommendation logic

Each of these goals was achieved and demonstrated in **Chapter 4: Results**, which showcased several test cases across different buyer profiles (e.g., first-time buyer, EV seeker, SUV family user). The outputs were consistently aligned with user expectations and logical prioritization.

5.4 Practical Relevance and Real-World Applicability

The system addresses real-world challenges faced by Indian consumers:

- Overwhelming choices across hundreds of models
- Uncertainty about hidden costs like maintenance
- Lack of transparency in feature-to-value comparison
- Complex EMI calculations for affordability estimation

By incorporating these concerns into the system design, the project offers more than a proof of concept—it provides a **foundational framework for future commercial tools** in the Indian auto-tech landscape.

5.5 Strengths of the Proposed System

- **Scalability**: Additional features (boot space, ADAS, resale value) can be added with minimal changes.
- **Explainability**: Final scores are derived from interpretable factors like mileage, sentiment, and service cost.
- **Responsiveness**: Real-time adaptability to user changes in budget or preferences.
- **Cost-Consciousness**: One of the few academic recommendation engines that model EMI limits and financial eligibility.

5.6 Reflections and Learning

During the development of this system, several insights were gained:

- **Real data is messy**: Sales data had to be cleaned post-OCR and validated manually.

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- **User behavior is complex**: Preferences are not static—what users say and what they choose may differ.
- **Integrating sentiment improves trust**: Including user reviews and sentiment built confidence in the recommendations.

This blend of **data science, UX sensitivity, and domain knowledge** was essential in creating a system that not only performs well technically but also resonates with real buyers.

5.7 Final Remarks

This project successfully demonstrates that a data-driven, sentiment-aware, financially viable recommendation system can **empower Indian car buyers with clarity and confidence**. It takes a step beyond generic filters and brings the user's priorities to the forefront of algorithmic intelligence.

By combining AI techniques with real-world decision parameters, this system lays the groundwork for a more human-centric approach to recommendation systems in the Indian automotive industry.

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CHAPTER 6

FUTURE SCOPE

6.1 Introduction

While the proposed car recommendation system has achieved its primary goals of personalization, affordability modeling, and sentiment-aware ranking, it also opens up several opportunities for further development and enhancement. As the automotive industry evolves and user behavior becomes increasingly data-driven, this system can be expanded to address emerging trends, market demands, and technical innovations.

This chapter outlines several potential directions in which the current system can be extended—ranging from algorithmic sophistication and data enrichment to full-stack deployment and real-world integration.

6.2 Integration with Real-Time Market Data

Currently, the system uses a static, cleaned dataset of specifications and reviews. In future versions:

- **APIs from automotive platforms** (e.g., CarDekho, CarWale, OLX Autos) can be integrated to fetch the latest price updates, discounts, real time user reviews and new car launches.
- Dynamic sales and pricing data will enable **real-time scoring**, improving the system's relevance in fast-changing markets.
- Real-time **fuel price data** by city can also be added to adjust fuel cost projection models for budget-sensitive users.

6.3 Expansion to Used Car Market

The current model focuses on new cars. However, **India's used car market is nearly 1.5x larger** than the new car segment and is growing rapidly. Future work could include:

- **Scraping used car listings** with real-time pricing and availability
- Adding condition-based scoring (e.g., kilometers run, accident history)
- Modeling **depreciation rates and resale value projections**
- Factoring **location-based availability** and delivery logistics

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6.4 Deep Learning-Based Ranking Models

While the current engine uses **cosine similarity and weighted scoring**, it can be extended to more sophisticated ML techniques:

- **Pairwise learning to rank** or **pointwise ranking models** using TensorFlow or PyTorch
- Integration of **XGBoost or LightGBM** for regression-based scoring
- Training models on **historical user interaction logs** to personalize based on implicit behavior
- Incorporation of **graph neural networks (GNNs)** to model car-user-brand relationships over time

6.5 Multi-Lingual NLP and Localized Sentiment Analysis

India is a linguistically diverse country. While the sentiment analysis is currently performed using DistilBERT on English reviews:

- Future systems can integrate **IndicNLP** or **MuRIL** for processing regional languages like Hindi, Tamil, Marathi, etc.
- **Code-mixed sentiment analysis** (e.g., Hinglish) will allow more accurate understanding of colloquial feedback.
- Regional language processing can enhance reach to **Tier 2 and Tier 3 cities**, expanding system inclusivity.

6.6 Voice-Based Input Interface

To enhance user accessibility:

- A **voice-based input system** can be implemented using Python's speech_recognition or Google's Speech-to-Text API.
- Users would be able to **speak preferences**, making the system useful for users who may not be comfortable typing or navigating interfaces.
- Especially useful for **mobile-first, semi-urban users** who are more comfortable with conversational interfaces.

6.7 Deployment as a Web Application

Though the current version runs in Jupyter, the system can be transitioned to a **web-based platform** using:

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- **Flask** or **Django** backend
- **React** or **Streamlit** frontend for interactivity
- Hosting on cloud platforms like **Heroku, AWS, or Azure**
- Integration with **mobile-first UIs** for better usability

This will make the tool accessible to a wider audience beyond academic or technical users.

6.8 Incorporation of Advanced Buyer Behavior Modeling

Future iterations can leverage **user analytics** to enhance personalization:

- **Session tracking** to detect behavior patterns
- Use of **A/B testing** to tune feature weights based on real engagement
- **Reinforcement learning** to adapt scores over time based on user feedback

6.9 Gamified User Feedback Loop

The system can introduce a **feedback and rating loop**, where users rate the relevance of their top 5 recommendations. This can be used to:

- Retrain weights dynamically
- Identify trends across demographics (e.g., millennials vs. families)
- Improve future predictions via supervised learning

Gamification through badges, comparisons, or savings tips can also increase engagement.

6.10 Summary

The proposed system, while complete in its current scope, has the potential to become a **feature-rich, production-ready recommendation platform**. It can evolve to include real-time data integration, deeper behavioral modeling, language diversity, and nationwide usability through voice and mobile support.

As India's automotive sector digitizes and consumer buying journeys become more informed and personalized, such systems will become essential. This project can serve as the foundation for a future-ready, AI-enabled vehicle selection assistant for the next generation of Indian car buyers.

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CHAPTER 7

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APPENDIX

Code 1 : Sentiment Analysis using DistilBERT

```
[ ] import nltk
    from nltk.sentiment import SentimentIntensityAnalyzer
    import pandas as pd

    nltk.download('vader_lexicon')
```

```
[nltk_data] Downloading package vader_lexicon to /root/nltk_data...
True
```

```
[ ] merged_df2 = merged_df2.rename(columns={'Review_Comment': 'Review'})
```

```
[ ] !pip install transformers
    !pip install torch

    import pandas as pd
    from transformers import pipeline

    classifier = pipeline("sentiment-analysis", model="distilbert-base-uncased-finetuned-sst-2-english")

    def analyze_sentiment(text):
        try:
            result = classifier(text)[0]
            return result['label']
        except Exception as e:
            print(f"Error analyzing sentiment for text '{text}': {e}")
            return None

    merged_df2['Sentiment'] = merged_df2['Review'].apply(analyze_sentiment)

    print(merged_df2.head())
    merged_df2.info()
```

Code 2 : Sales Data Clean Preprocessing and Trend Score

```
import pandas as pd
import os
import glob

def combine_csv_columns(folder_path, output_excel_path):
    all_files = glob.glob(os.path.join(folder_path, "*.csv"))
    df_list = []

    for filename in all_files:
        try:
            df = pd.read_csv(filename, index_col=None, header=0)
            df_list.append(df)
        except pd.errors.EmptyDataError:
            print(f"Warning: Skipping empty file {filename}")
        except pd.errors.ParserError:
            print(f"Warning: Skipping file with parsing error {filename}")

    if not df_list:
        print("Error: No valid CSV files found in the specified directory.")
        return

    combined_df = pd.concat(df_list, axis=1)
    combined_df.to_excel(output_excel_path, index=False)
    print(f"Combined CSV files saved to {output_excel_path}")

folder_path = '/content/drive/MyDrive/Sem6-Project/sales_data_csv'
output_excel_path = '/content/drive/MyDrive/Sem6-Project/sales_data_csv/combined_data.xlsx'

os.makedirs(folder_path, exist_ok=True)

import numpy as np
for i in range(3):
  data = np.random.rand(10, 2)
  df = pd.DataFrame(data, columns=[f'Col{i*2}', f'Col{i*2 + 1}'])
  df.to_csv(os.path.join(folder_path, f'file_{i}.csv'), index=False)

combine_csv_columns(folder_path, output_excel_path)
```


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```python
import pandas as pd
import numpy as np

df = pd.read_excel('/content/drive/MyDrive/Sem6-Project/Sales_Data/Restructured_Car_Sales_Sorted_2021_2024_v1.xlsx')

df.fillna(0, inplace=True)

def calculate_trend_score(row):
    sales_data = row[2:].values
    year_2023 = sales_data[24:36]
    year_2024 = sales_data[36:]
    if sum(year_2023) == 0 and sum(year_2024) == 0:
        return 0
    elif sum(year_2023) > 0 and len(year_2024) >= 6 and sum(year_2024[6:]) == 0:
        return 0

    total_2023 = sum(year_2023)
    total_2024 = sum(year_2024)

    avg_2023 = total_2023 / 12 if len(year_2023) == 12 else 0
    avg_2024 = total_2024 / len(year_2024) if len(year_2024) > 0 else 0

    if len(year_2024) > 0 and avg_2024 < 500:
        return 1

    if avg_2023 > 0 and len(year_2024) > 0:
        growth_rate = (avg_2024 - avg_2023) / avg_2023 * 100
    else:
        growth_rate = 0

    momentum = 0
    if len(year_2023) >= 12 and len(year_2024) >= 6:
        first_half_2023 = sum(year_2023[:6])
        last_half_2023 = sum(year_2023[6:12])
        first_half_2024 = sum(year_2024[:6])
```

```python
        momentum_2023 = (last_half_2023 - first_half_2023) / first_half_2023 * 100 if first_half_2023 > 0 else 0
        momentum_2024 = (first_half_2024 - last_half_2023) / last_half_2023 * 100 if last_half_2023 > 0 else 0
        momentum = (momentum_2023 + momentum_2024) / 2

    score = 3

    if growth_rate > 30:
        score += 1
    elif growth_rate > 15:
        score += 0.5
    elif growth_rate < -30:
        score -= 1
    elif growth_rate < -15:
        score -= 0.5

    if momentum > 20:
        score += 1
    elif momentum > 10:
        score += 0.5
    elif momentum < -20:
        score -= 1
    elif momentum < -10:
        score -= 0.5

    if total_2023 == 0 and total_2024 > 0:
        score = 5 if avg_2024 > 1000 else 4

    return max(1, min(5, round(score)))

df['Trend Score'] = df.apply(calculate_trend_score, axis=1)
df['Discontinued'] = df.apply(lambda x: calculate_trend_score(x) == 0, axis=1)
df['Significant Decline'] = df.apply(lambda x: calculate_trend_score(x) == 1 and not x['Discontinued'], axis=1)

df.to_excel('Car_Sales_Final_Trend_Analysis.xlsx', index=False)
print("Analysis complete. 0=Discontinued, 1=Severe Decline, 2-5=Active models")
```


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Code 3 : Top 5 Recommendation

```
import pandas as pd
import numpy as np
from sklearn.metrics.pairwise import cosine_similarity
from sklearn.preprocessing import MinMaxScaler
from sklearn.linear_model import SGDRegressor
from scipy import stats
import ipywidgets as widgets
from IPython.display import display, HTML
import warnings
import pickle
import os
import time
warnings.filterwarnings("ignore")

df = pd.read_excel("/content/drive/MyDrive/Sem6-Project/Final_Datasets/Indian_Cars_Data_25Mar_v1.xlsx")

df1 = df[df['Price'] <= 5000000]
df1 = df1.dropna(subset=['Seating_Capacity'])
makes_to_keep = ['Maruti', 'Hyundai', 'Honda', 'Tata', 'Mahindra', 'Toyota','Kia', 'Skoda', 'Volkswagen', 'Renault', 'Nissan', 'FCA', 'Citroen', 'MG']
df1 = df1[df1['Make'].isin(makes_to_keep)]
df1 = df1.drop_duplicates(subset=['Make', 'Model', 'Price'])
df1['Sales_Trend'].fillna(2, inplace=True)
df1['Car_Wale_Score'].fillna(1, inplace=True)
df1['Car_Trade_Score'].fillna(1, inplace=True)
df1['Service_Cost (5 Yrs)'] = pd.to_numeric(df1['Service_Cost (5 Yrs)'], errors='coerce')
df1['Service_Cost (5 Yrs)'] = df1.groupby('Make')['Service_Cost (5 Yrs)'].transform(lambda x: x.fillna(x.max()))
df1['Transmission'] = df1['Transmission'].fillna('Manual')
df1.loc[df1['Variant'].str.contains('Zxi AMT', na=False), 'Transmission'] = 'AMT'
df1['Mileage (Km/L)'] = df1.groupby('Make')['Mileage (Km/L)'].transform(lambda x: x.fillna(x.mean()))
df1.loc[df1['Fuel Type'] == 'Electric', 'Mileage (Km/L)'] = np.nan
df1['Combined_Score'] = (df1['Car_Wale_Score'] + df1['Car_Trade_Score']) / 2

scaler = MinMaxScaler()
features_to_scale = ['Mileage (Km/L)', 'Combined_Score', 'Service_Cost (5 Yrs)', 'Sales_Trend']
df1[features_to_scale] = scaler.fit_transform(df1[features_to_scale])
```

```
content_features = df1[['Mileage (Km/L)', 'Combined_Score', 'Service_Cost (5 Yrs)', 'Sales_Trend']]
content_features = content_features.dropna()
content_similarity_matrix = cosine_similarity(content_features)

class CarRecommender:
    def __init__(self, data):
        self.df = data
        self.user_profile = None
        self.weights = {
            'mileage': 4.29,
            'combined_scores': 4.36,
            'preferred_make': 4.07,
            'service_cost': 3.61,
            'sales_trend': 2.55,
            'current_car_brand': 2.13
        }
        self.normalize_weights()
        self._prepare_similarity_matrix()
        self.feedback_model = None
        self.load_feedback_model()

    def normalize_weights(self):
        total = sum(self.weights.values())
        for key in self.weights:
            self.weights[key] = self.weights[key] / total

    def _prepare_similarity_matrix(self):
        features_for_similarity = ['Mileage (Km/L)', 'Combined_Score',
                                   'Service_Cost (5 Yrs)', 'Sales_Trend']
        features_df = self.df[features_for_similarity].dropna()
        self.content_similarity_matrix = cosine_similarity(features_df)
        self.valid_indices = features_df.index
```


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```python
def load_feedback_model(self):
    model_path = 'feedback_model.pkl'
    if os.path.exists(model_path):
        with open(model_path, 'rb') as f:
            self.feedback_model = pickle.load(f)
    else:
        self.feedback_model = SGDRegressor()
        X = np.array(list(self.weights.values())).reshape(-1, 1)
        y = np.ones(X.shape[0])
        self.feedback_model.fit(X, y)

def save_feedback_model(self):
    with open('feedback_model.pkl', 'wb') as f:
        pickle.dump(self.feedback_model, f)

def update_weights_based_on_feedback(self, feedback_scores):

    X = np.array(list(self.weights.values())).reshape(-1, 1)
    y = np.array([np.mean(feedback_scores)])

    self.feedback_model.partial_fit(X, y)

    new_weights = self.feedback_model.coef_.flatten()

    total = sum(new_weights)
    for i, key in enumerate(self.weights.keys()):
        self.weights[key] = new_weights[i] / total

    self.save_feedback_model()
```

```python
def create_user_profile(self, preferences):
    self.user_profile = {
        'preferred_make': preferences.get('preferred_make', None),
        'current_car_brand': preferences.get('current_car_brand', None),
        'budget_range': (preferences.get('min_budget', 0),
                         preferences.get('max_budget', float('inf'))),
        'seating': preferences.get('seating', None),
        'fuel_type': preferences.get('fuel_type', None),
        'transmission': preferences.get('transmission', None),
        'financial_info': {
            'net_income': preferences.get('net_income', None),
            'monthly_commitments': preferences.get('monthly_commitments', None),
            'down_payment': preferences.get('down_payment', None)
        }
    }

def car_loan_eligibility(self, car_price):
    if not self.user_profile['financial_info']['net_income']:
        return "Not Checked"

    monthly_income = self.user_profile['financial_info']['net_income']
    monthly_commitments = self.user_profile['financial_info']['monthly_commitments']
    down_payment = self.user_profile['financial_info']['down_payment']

    disposable_income = monthly_income - monthly_commitments
    max_affordable_emi = disposable_income * 0.5

    loan_amount = car_price - down_payment
    tenure = 60
    interest_rate = 10

    monthly_rate = interest_rate / 12 / 100
    emi = (loan_amount * monthly_rate * (1 + monthly_rate)**tenure) / ((1 + monthly_rate)**tenure - 1)

    ltv = (loan_amount / car_price) * 100
```


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```python
        if emi <= max_affordable_emi and 30 <= ltv <= 85:
            return "Yes"
        return "No"

    def content_based_filtering(self):
        filtered = self.df[
            (self.df['Price'] >= self.user_profile['budget_range'][0]) &
            (self.df['Price'] <= self.user_profile['budget_range'][1])
        ].copy()

        if self.user_profile['seating']:
            filtered = filtered[filtered['Seating_Capacity'] == self.user_profile['seating']]
        if self.user_profile['fuel_type'] and self.user_profile['fuel_type'] != 'Any':
            filtered = filtered[filtered['Fuel Type'] == self.user_profile['fuel_type']]
        if self.user_profile['transmission'] and self.user_profile['transmission'] != 'Any':
            filtered = filtered[filtered['Transmission'] == self.user_profile['transmission']]

        if len(filtered) == 0:
            return None

        filtered = filtered.reset_index(drop=True)
        valid_filtered = filtered[filtered.index.isin(self.valid_indices)]
        original_indices = valid_filtered.index.tolist()

        similarity_scores = []
        for idx in original_indices:
            if idx < len(self.content_similarity_matrix):
                similarity_scores.append(self.content_similarity_matrix[idx].mean())
            else:
                similarity_scores.append(0)

        valid_filtered['Combined_Scores'] = similarity_scores
```

```python
        if len(valid_filtered) < len(filtered):
            make_median_scores = valid_filtered.groupby('Make')['Combined_Scores'].median()
            filtered['Combined_Scores'] = filtered.merge(make_median_scores, on='Make', how='left')['Combined_Scores']

        else:
            filtered = valid_filtered

        filtered = filtered.sort_values(['Make', 'Model', 'Price'])
        median_filtered = (
            filtered.groupby(['Make', 'Model'], group_keys=False)
            .apply(lambda x: x.iloc[len(x)//2] if len(x) > 0 else None)
            .reset_index(drop=True)
        )

        return median_filtered

    def calculate_weighted_score(self, row):
        mileage_score = row['Mileage (Km/L)'] if pd.notna(row['Mileage (Km/L)']) else 0.5
        combined_scores = row.get('Combined_Scores', 0)
        preferred_make_score = 1 if (self.user_profile['preferred_make'] and
                                     row['Make'] == self.user_profile['preferred_make']) else 0
        service_score = 1 - row['Service_Cost (5 Yrs)'] if pd.notna(row['Service_Cost (5 Yrs)']) else 0.5
        sales_score = row['Sales_Trend'] if pd.notna(row['Sales_Trend']) else 0.5
        current_brand_score = 1 if (self.user_profile['current_car_brand'] and
                                     row['Make'] == self.user_profile['current_car_brand']) else 0

        final_score = (
            mileage_score * self.weights['mileage'] +
            combined_scores * self.weights['combined_scores'] +
            preferred_make_score * self.weights['preferred_make'] +
            service_score * self.weights['service_cost'] +
            sales_score * self.weights['sales_trend'] +
            current_brand_score * self.weights['current_car_brand']
        )

        return final_score
```


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```python
    def recommend(self, preferences):
        self.create_user_profile(preferences)
        filtered = self.content_based_filtering()

        if filtered is None or len(filtered) == 0:
            return None

        filtered['Final_Score'] = filtered.apply(self.calculate_weighted_score, axis=1)

        if self.user_profile['financial_info']['net_income']:
            filtered['Financial_Eligibility'] = filtered['Price'].apply(
                lambda x: self.car_loan_eligibility(x))
        else:
            filtered['Financial_Eligibility'] = "Not Checked"

        return filtered.sort_values('Final_Score', ascending=False)

def create_widgets():
    makes = ['Any'] + sorted(df1['Make'].unique().tolist())
    fuel_types = ['Any'] + sorted(df1['Fuel Type'].unique().tolist())
    transmissions = ['Any'] + sorted(df1['Transmission'].unique().tolist())
    seating_options = ['Any'] + sorted(df1['Seating_Capacity'].dropna().unique().astype(int).tolist())

    widgets_dict = {
        'preferred_make': widgets.Dropdown(options=makes, value='Any', description='Preferred Make:'),
        'current_brand': widgets.Dropdown(options=makes, value='Any', description='Current Brand:'),
        'min_budget': widgets.IntSlider(value=500000, min=0, max=5000000, step=100000, description='Min Budget:'),
        'max_budget': widgets.IntSlider(value=3000000, min=0, max=5000000, step=100000, description='Max Budget:'),
        'seating': widgets.Dropdown(options=seating_options, value=5, description='Seating:'),
        'fuel_type': widgets.Dropdown(options=fuel_types, value='Petrol', description='Fuel Type:'),
        'transmission': widgets.Dropdown(options=transmissions, value='Manual', description='Transmission:'),
        'financial_assessment': widgets.Dropdown(options=['Yes', 'No'], value='No', description='Financial Check:'),
        'net_income': widgets.FloatText(value=1.0, description='Net Income (Lakhs/month):'),
        'monthly_commitments': widgets.FloatText(value=0.5, description='Monthly Commitments (Lakhs):'),
```

```python
        'down_payment': widgets.FloatText(value=0.1, description='Down Payment (Lakhs):'),
        'submit_button': widgets.Button(description='Get Recommendations', button_style='success')
    }

    widgets_dict['net_income'].layout.visibility = 'hidden'
    widgets_dict['monthly_commitments'].layout.visibility = 'hidden'
    widgets_dict['down_payment'].layout.visibility = 'hidden'

    for w in widgets_dict.values():
        w.style.description_width = 'initial'

    return widgets_dict

def on_financial_change(change, widgets):
    if change['new'] == 'Yes':
        widgets['net_income'].layout.visibility = 'visible'
        widgets['monthly_commitments'].layout.visibility = 'visible'
        widgets['down_payment'].layout.visibility = 'visible'
    else:
        widgets['net_income'].layout.visibility = 'hidden'
        widgets['monthly_commitments'].layout.visibility = 'hidden'
        widgets['down_payment'].layout.visibility = 'hidden'

def collect_feedback(recommendations, recommender, preferences):
    display(HTML("<h3>Please rate these recommendations (1-5):</h3>"))
    feedback_widgets = []
    slider = widgets.IntSlider(min=1, max=5, value=3, description="Overall Feedback")
    display(slider)
    feedback_widgets.append(slider)

    submit_button = widgets.Button(description="Submit Feedback", button_style='info')
    display(submit_button)
```


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```python
    def on_feedback_submit(b):
        feedback_scores = [w.value for w in feedback_widgets]
        display(HTML("<p style='color:green'>Thank you for your feedback!</p>"))

        evaluation = evaluate_recommendations(recommendations, preferences, feedback_scores)

        display(HTML("<h3>Recommendation Evaluation</h3>"))
        for metric, value in evaluation.items():
            display(HTML(f"<p><b>{metric}:</b> {value}</p>"))

    submit_button.on_click(on_feedback_submit)


def evaluate_recommendations(recommendations, user_inputs, feedback_scores):

    top_recommendations = recommendations.head(5).copy()

    USS = (feedback_scores[0] / 5) * 100

    budget_mid = (user_inputs['min_budget'] + user_inputs['max_budget']) / 2
    top_recommendations['Price Deviation'] = abs(top_recommendations['Price'] - budget_mid) / budget_mid * 100
    PDS = top_recommendations['Price Deviation'].mean()

    match_count = 0
    for _, row in recommendations.iterrows():
        match_count += (row['Make'] == user_inputs['preferred_make']) + \
                       (row['Fuel Type'] == user_inputs['fuel_type']) + \
                       (row['Transmission'] == user_inputs['transmission'])

    PMR = (match_count / (len(top_recommendations) * 3)) * 100

    if 'Financial_Eligibility' in top_recommendations:
        FFS = (top_recommendations['Financial_Eligibility'].value_counts(normalize=True).get('Yes', 0)) * 100
    else:
        FFS = 0
```


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```python
    if len(top_recommendations) > 1:
        final_scores = top_recommendations['Final_Score'].values
        cv = np.std(final_scores) / np.mean(final_scores)
        WSC = 1 - cv
    else:
        WSC = 0



    evaluation_results = {
        "User Satisfaction Score (USS)": f"{USS:.2f}%",
        "Price Deviation Score (PDS)": f"{PDS:.2f}%",
        "Preference Match Rate (PMR)": f"{PMR:.2f}%",
        "Financial Feasibility Score (FFS)": f"{FFS:.2f}%",
        "Weighted Score Consistency (WSC)": f"{WSC:.2f}",
    }

    return evaluation_results

def main():
    widgets_dict = create_widgets()
    recommender = CarRecommender(df1)

    for w in widgets_dict.values():
        if w != widgets_dict['submit_button']:
            display(w)

    widgets_dict['financial_assessment'].observe(
        lambda change: on_financial_change(change, widgets_dict),
        names='value'
    )

    display(widgets_dict['submit_button'])
```

```python
def on_submit(b):

    preferences = {
        'preferred_make': widgets_dict['preferred_make'].value if widgets_dict['preferred_make'].value != 'Any' else None,
        'current_car_brand': widgets_dict['current_brand'].value if widgets_dict['current_brand'].value != 'Any' else None,
        'min_budget': widgets_dict['min_budget'].value,
        'max_budget': widgets_dict['max_budget'].value,
        'seating': int(widgets_dict['seating'].value),
        'fuel_type': widgets_dict['fuel_type'].value if widgets_dict['fuel_type'].value != 'Any' else None,
        'transmission': widgets_dict['transmission'].value if widgets_dict['transmission'].value != 'Any' else None,
        'net_income': widgets_dict['net_income'].value * 100000 if widgets_dict['financial_assessment'].value == 'Yes' else None,
        'monthly_commitments': widgets_dict['monthly_commitments'].value * 100000 if widgets_dict['financial_assessment'].value == 'Yes' else None,
        'down_payment': widgets_dict['down_payment'].value * 100000 if widgets_dict['financial_assessment'].value == 'Yes' else None
    }

    start_time = time.time()
    recommendations = recommender.recommend(preferences)
    end_time = time.time()
    response_time = end_time - start_time
    print(f"Response time: {response_time:.4f} seconds")

    if recommendations is None or len(recommendations) == 0:
        display(HTML("<p style='color:red'>No cars match your criteria. Please adjust your filters.</p>"))
        return

    top_recommendations = recommendations.head(5).copy()
    top_recommendations['Price (Lakhs)'] = top_recommendations['Price'] / 100000

    service_cost_mapping = df.set_index(['Make', 'Model'])['Service_Cost (5 Yrs)'].to_dict()

    def get_actual_service_cost(row):
        return service_cost_mapping.get((row['Make'], row['Model']))

    top_recommendations['Service_Cost (5 Yrs)'] = top_recommendations.apply(get_actual_service_cost, axis=1)
```


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```
        display_cols = ['Make', 'Model', 'Price (Lakhs)', 'Fuel Type', 'Transmission',
                        'Seating_Capacity', 'Service_Cost (5 Yrs)', 'Final_Score', 'Financial_Eligibility']

        display(HTML("<h3>Top 5 Car Recommendations</h3>"))
        display(top_recommendations[display_cols].style.format({
            'Price (Lakhs)': '₹{:.2f}',
            'Seating_Capacity': '{:.0f}',
            'Final_Score': '{:.2f}',
            'Service_Cost (5 Yrs)': '₹{:.0f}'
        }))

        collect_feedback(top_recommendations, recommender, preferences)

    widgets_dict['submit_button'].on_click(on_submit)

    def on_submit_feedback(button):
        feedback_scores = [float(w.value) for w in feedback_widgets]
        system.update_weights_based_on_feedback(feedback_scores)

if __name__ == "__main__":
    main()
```


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Abstract:
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